

TFT LCD 구동 IC의 설계

Dec. 2004

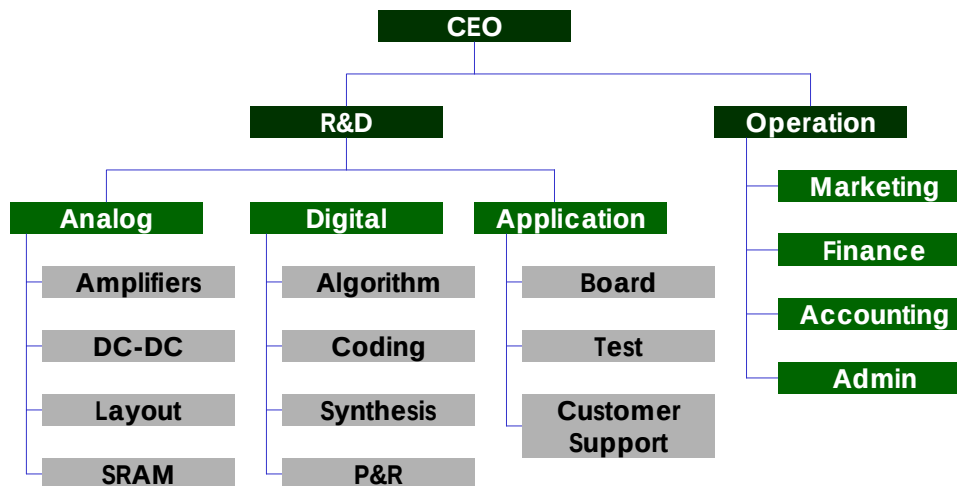
김 관
Pixel Chips Co., Ltd.

02-552-9428
kimkwan@pixelchips.com

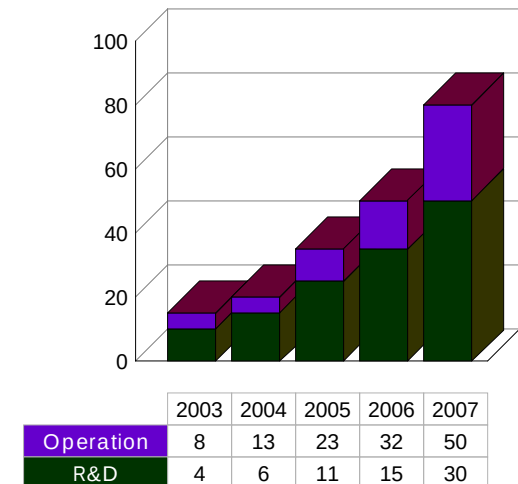
Company Overview

Foundation	Mar. 3, 2000
Equity Financing	Around USD 3,500,000
Common Stock	Around USD 1,300,000
No. of Employees	16 (11 R&D Engineers)
Products	TFT LCD One-Chip Driver IC's for Mobile Phones

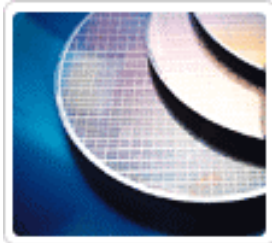
Organization



HR Projection



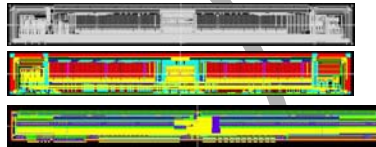
Fab Foundry



Silterra

Magnachip

LCD Driver IC



Determines the LCD driving scheme and the module architecture. Interprets the commands from MPU and follows the command to drive the LCD. Many kinds of LCD driving modes and configurations are supported.

Application Support

Supports the customers to set the LCD driving configuration and to optimize the module performance.

LCD Module



TFT LCD module with one-chip driver



Mobile Phone

260,000 color, 1.8" ~ 2.4" TFT-LCD for mobile phones

Standard Compliancy of Our Products

132x176	with memory	Developed (2 kinds of IC's)
176x240	with memory	Under Development
240x320	gateless or memoryless	Under Planning
320x480 or VGA	gateless and 3-mux source	Under Planning

260,000 color, one-chip drivers

Mobile TFT Infrastructure

Emerging Market
Mobile Phone, 2.5G, 3G

WW TFT Foundry Capa.
Release
G3, G3.5 release cells

PixelChips
Mobile TFT Solution Provider

Competing Solution
Providers
Hitachi, Sharp, NEC, Samsung

WW Assembly Capa.
Support
Taiwan, Korea, China

Partners (Production Line)

FAB Foundry



0.18um/30V, 0.25um, 0.35um/18V,
0.60um/40V
Bump/Test/RT turnkey service



0.20um/30V

Bumping



CP (Wafer-Level Test)



Dicing & P&P



TCP/COF



RT/ESD (Over-Stress Experiment)



FT (Package-Level Test)



Backend turnkey service

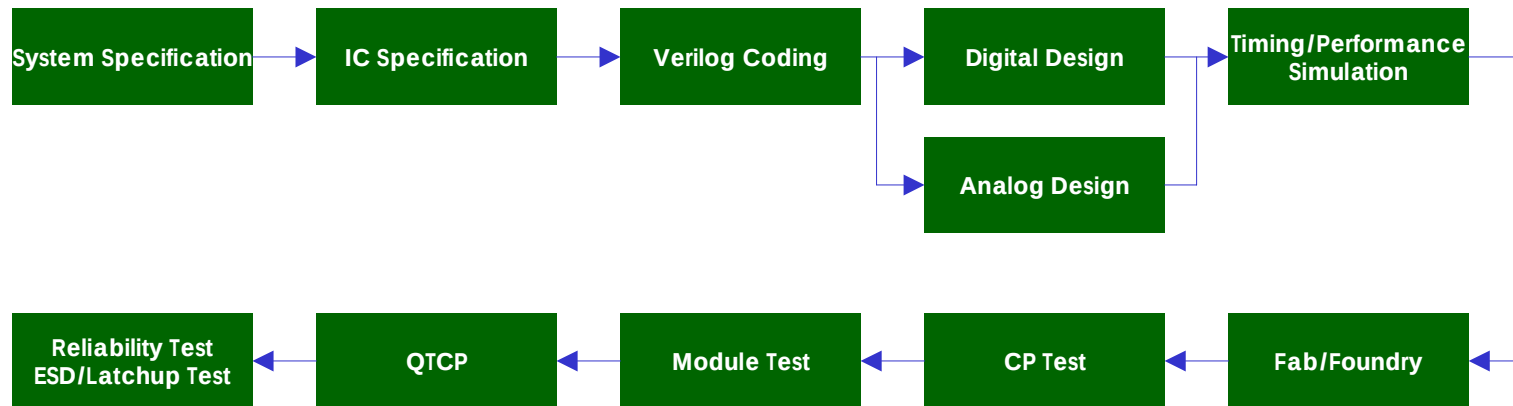


King Yuan ELECTRONICS CO., LTD.



SRC Semiconductor

Development Phase



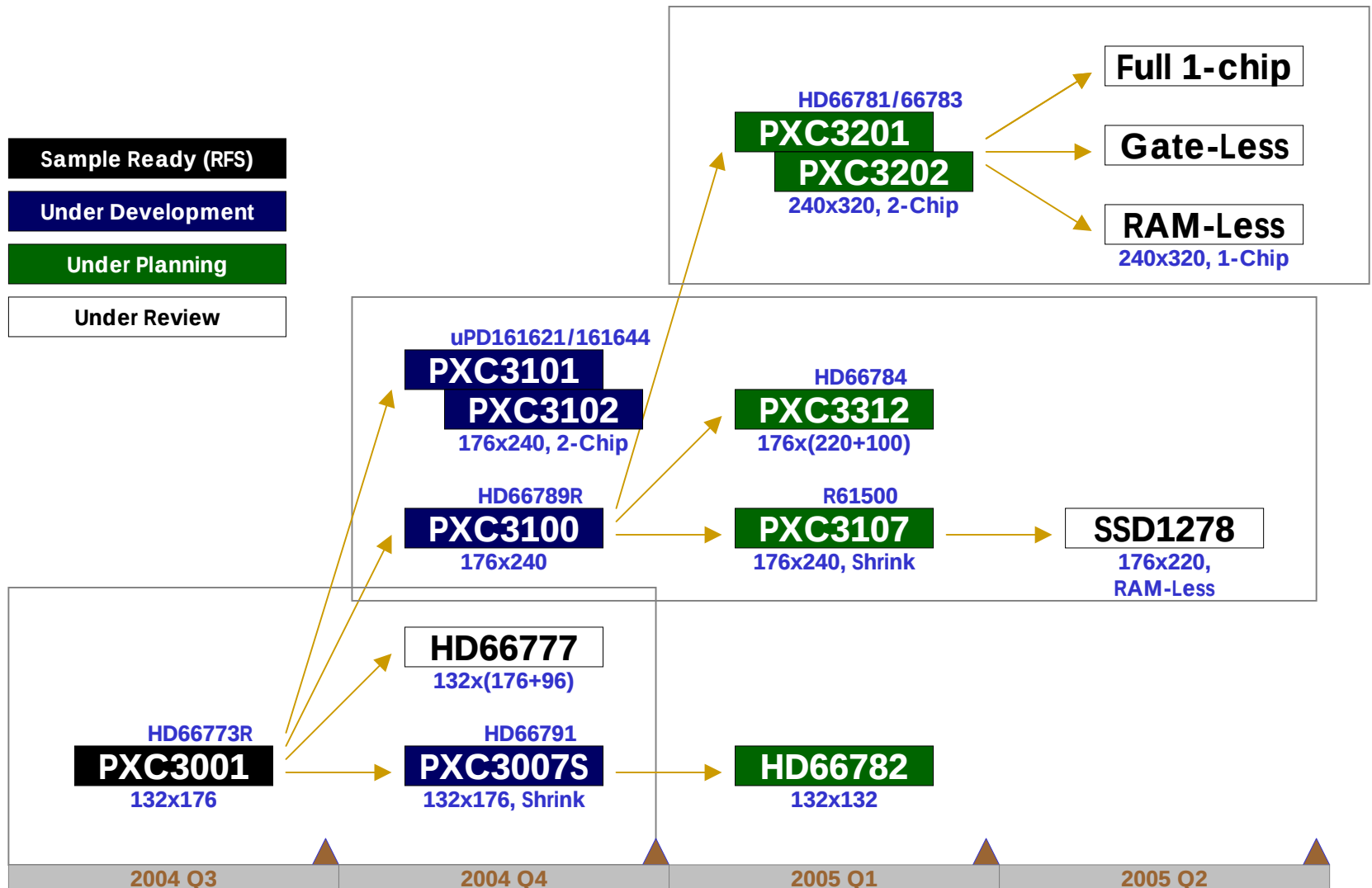
Commercialization Phase

(together with module makers)

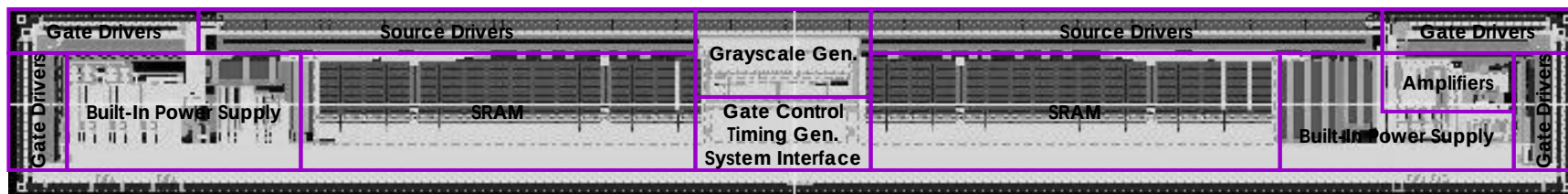


On-Site Engineering Support

Product Portfolio



PXC3002S Complexity

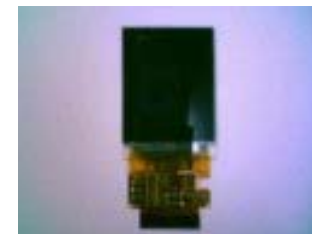


Process: 0.20um, 2.5V/5.0V/32V.
Die Size: 20,660um x 2,440um with scribe lane

One-chip display driver

Source drivers + gate drivers + frame memory + display timing controller + system interface + grayscale generator amplifiers + built-in DC-DC converters + voltage followers

Only some passive elements (R/C) and some diodes are added on the FPC.



Mobile Phone TFT LDI Competition



Major contributor of mobile phone TFT LDI's.
Leads the market standards.
Provides a variety of product series.



Pixelchips' Strategy:
Lower cost and better engineering support



Taiwan local IC vendor.
CMO is a shareholder of Himax.



Another Taiwan local vendor.
UMC is a shareholder of Novatek.



Pixelchips' Strategy:
Earlier product release and better quality



Samsung LDI has a big in-house customer.
Goes to the gate on glass configuration.

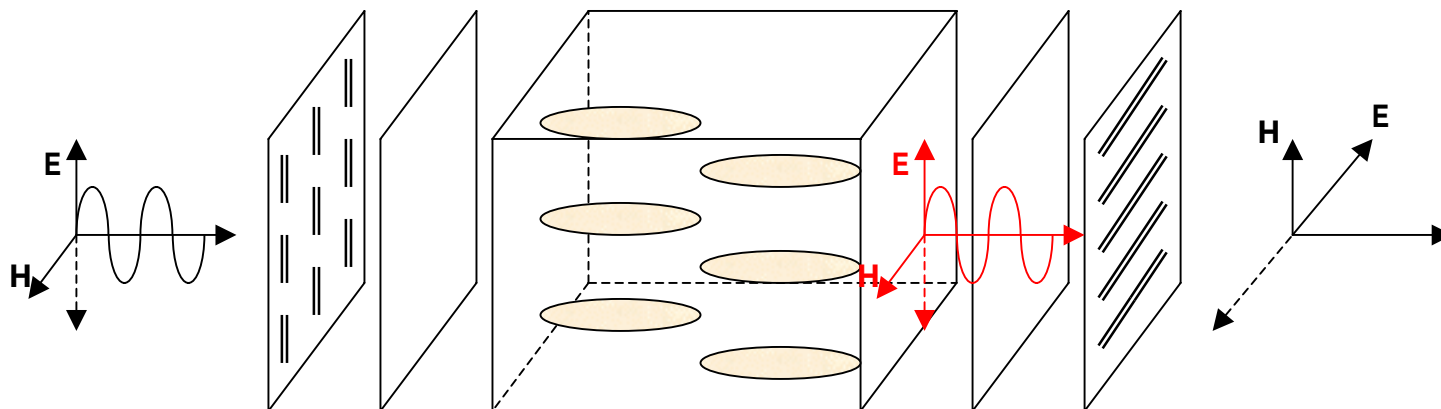
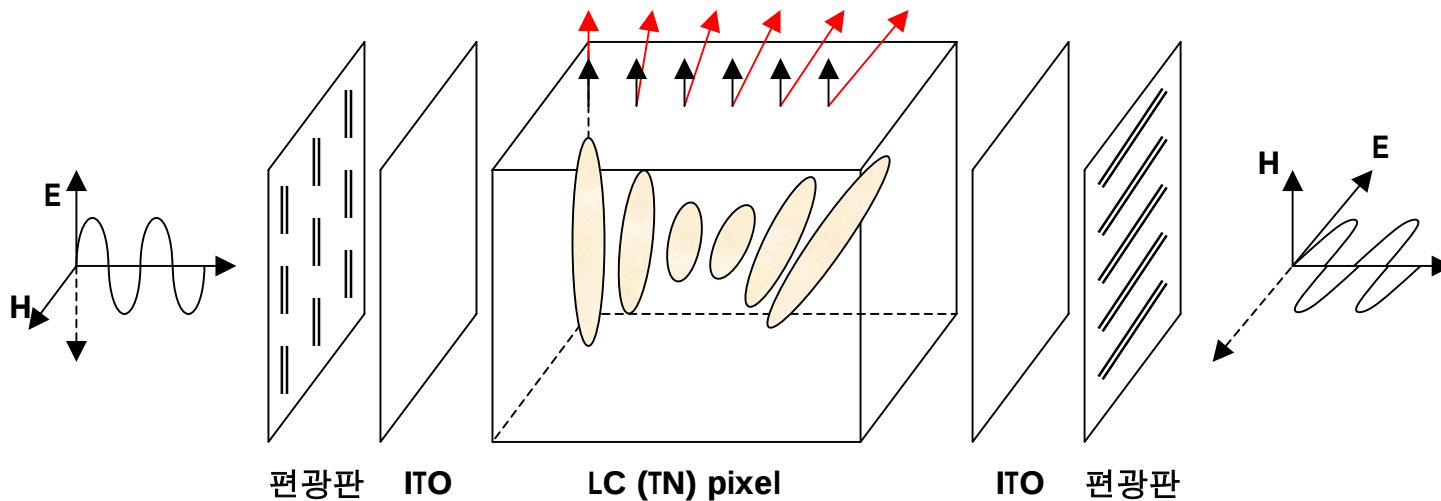


Pixelchips' Strategy:
Lower cost and earliest as a 2nd vendor



Provides a full series of mobile phone LDI.

Liquid Crystal Display

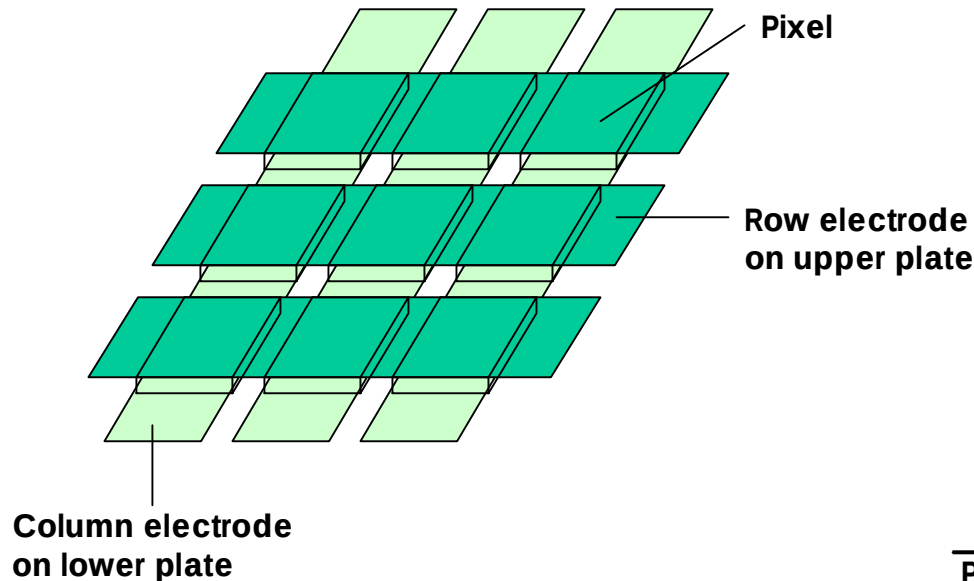


Normally white

Normally black ← 두 개의 편광판 방향이 같다면..

LC 배열의 조절 mode → LCD의 종류

Passive Matrix LCD



과거의 흑백 STN notebook LCD

현재의 흑백/color 휴대폰

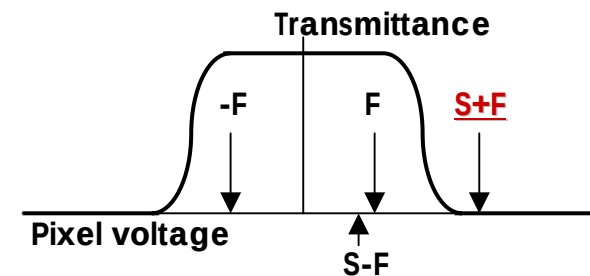
4가지의 가능한 pixel 전압 조합으로 pixel 단위로 ON/OFF 조절.

$$V_{\text{pixel}} = V_{\text{row}} - V_{\text{column}}$$

$$V_{\text{row}} = \{0, S\}$$

$$V_{\text{column}} = \{F, -F\}$$

$$V_{\text{pixel}} = \{S+F, F, S-F, -F\}$$

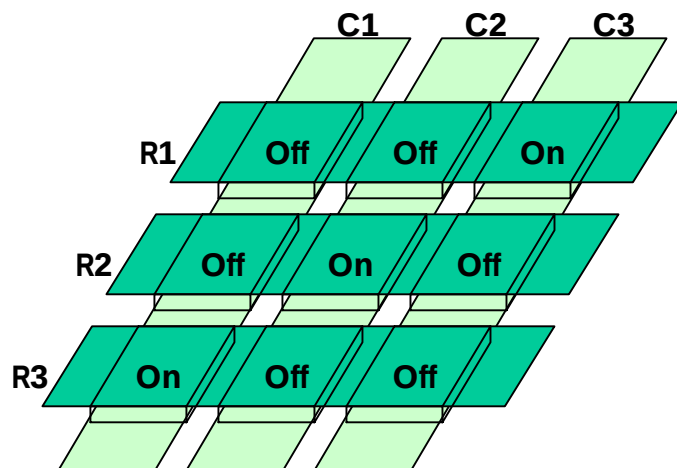


STN LCD

270도 twisting

ON/OFF transition이 급격하다.

Passive Matrix Addressing

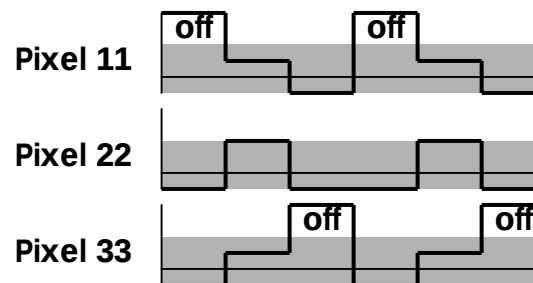
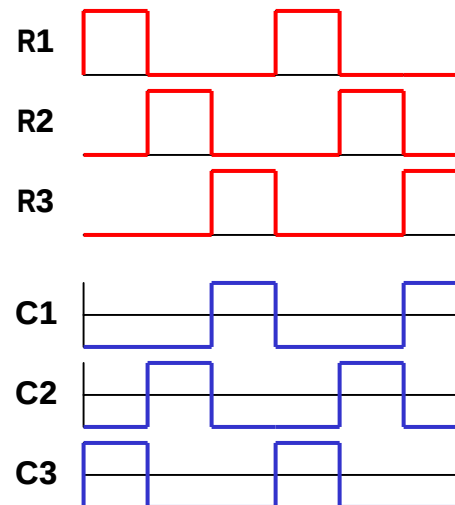


유효 표시 시간 = $1/(\# \text{ Lines})$

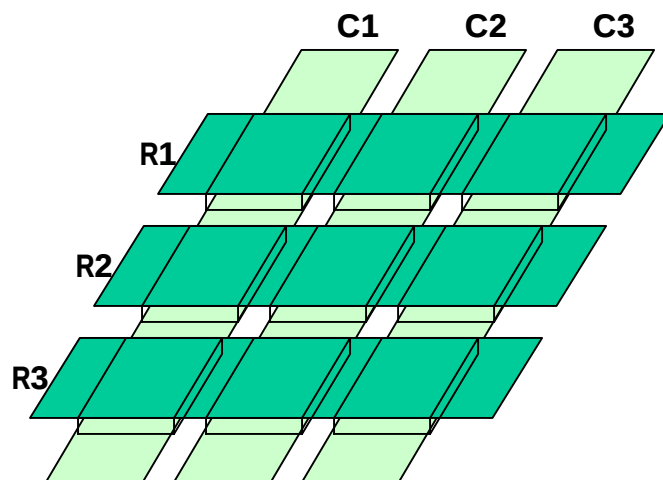
시간 평균 rms → 밝기 결정

LCD 응답 속도 > 150 ms

신호 대비 비율이 낮다.



PMLCD with the Polarity Inversion

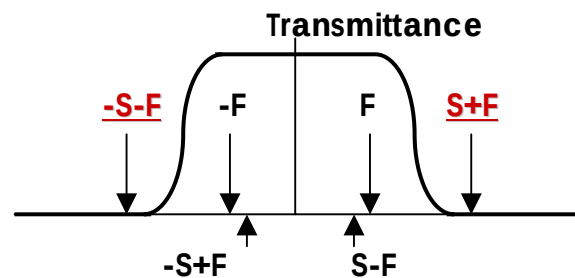


$$V_{\text{pixel}} = V_{\text{row}} - V_{\text{column}}$$

$$V_{\text{row}} = \{S, 0, -S\}$$

$$V_{\text{column}} = \{F, -F\}$$

$$V_{\text{pixel}} = \{\underline{S+F}, F, S-F, -S+F, -F, \underline{-S-F}\}$$



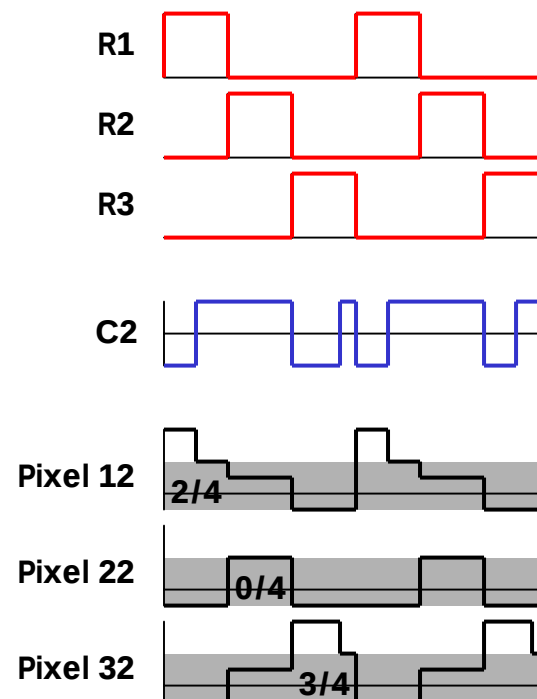
LC 수명 문제

Multi-Gray Level PMLCD

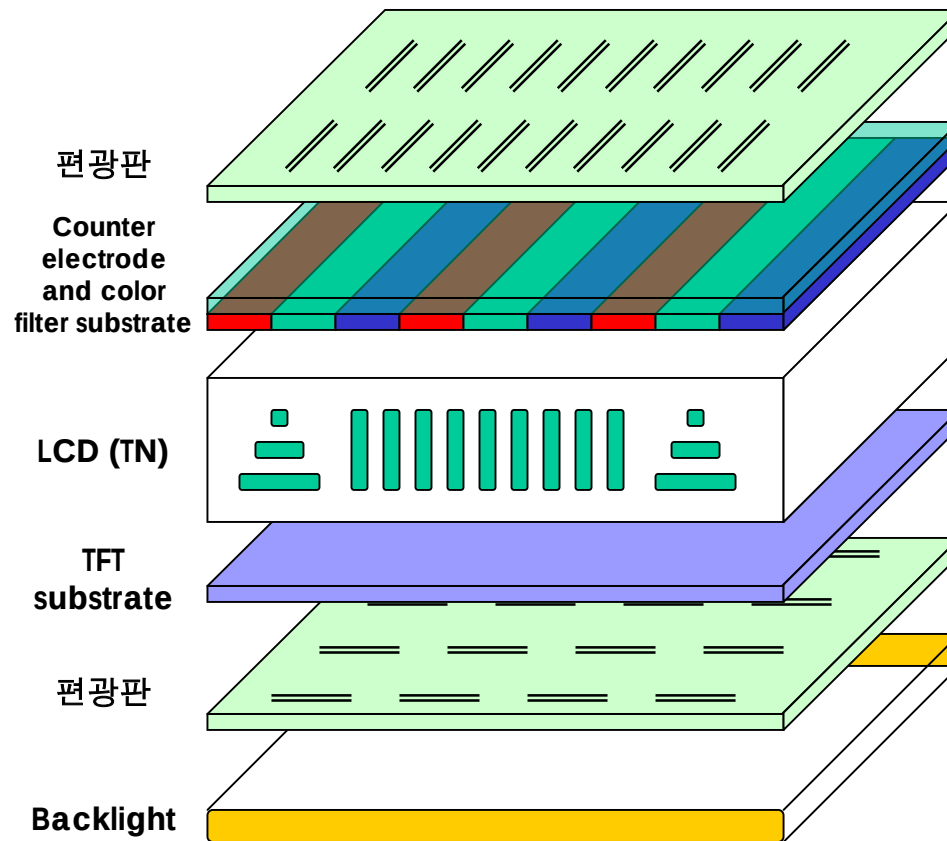
Frame Rate Control (FRC)

Frame	1st	2nd	3rd	4th	Result	
Pixel 1					=	 5/5 grey
Pixel 2					=	 4/5 grey
Pixel 3					=	 3/5 grey
Pixel 4					=	 2/5 grey
Pixel 5					=	 1/5 off

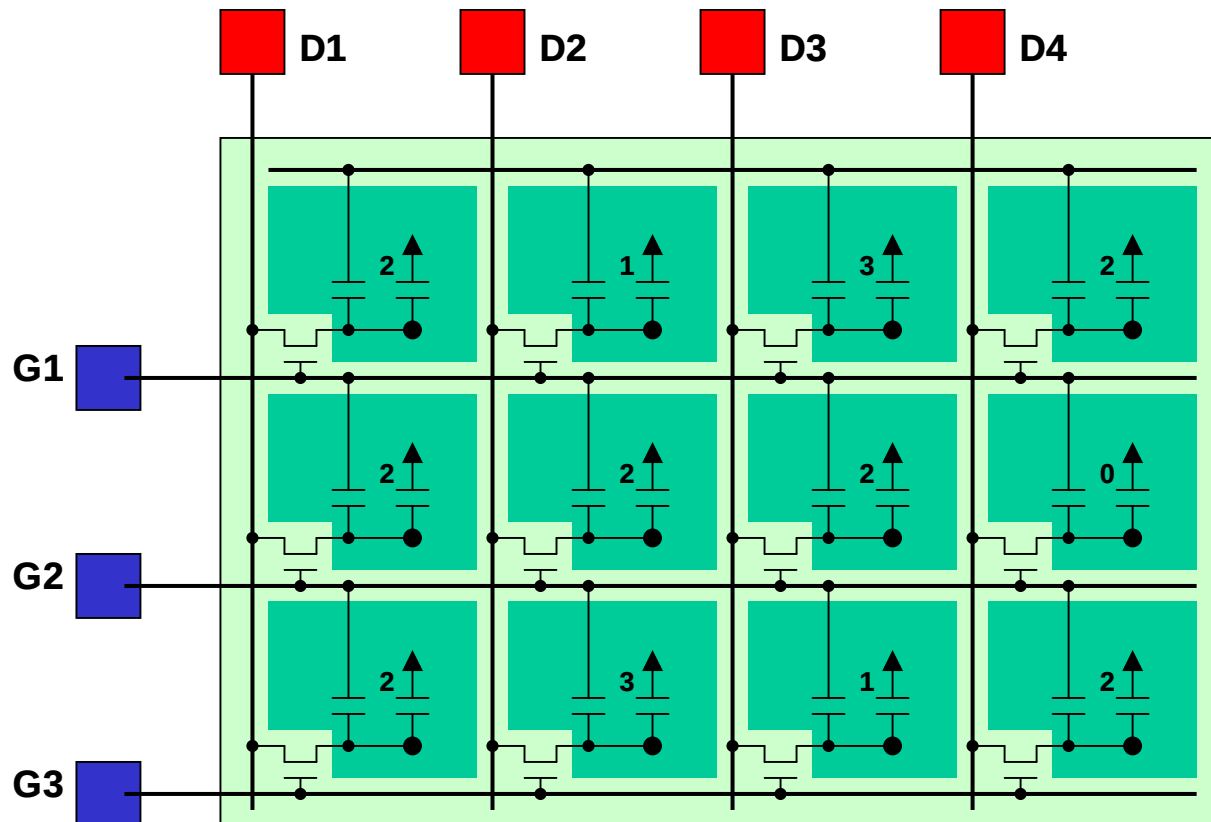
Pulse Width Modulation (PWM)



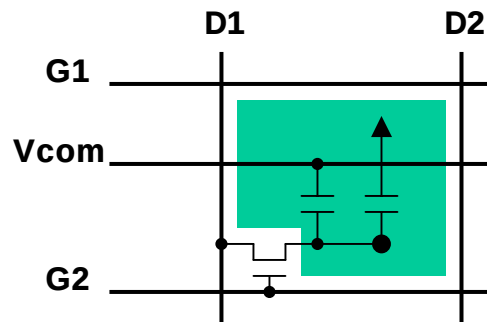
Color TFT LCD의 구조



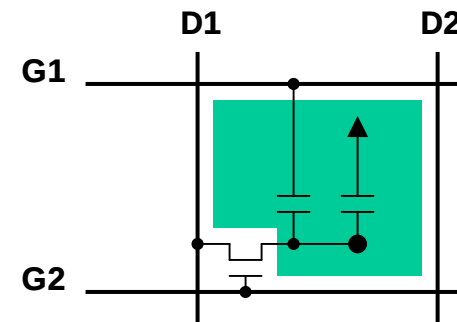
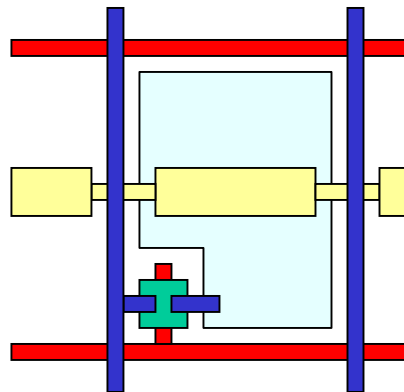
TFT 기판의 구조



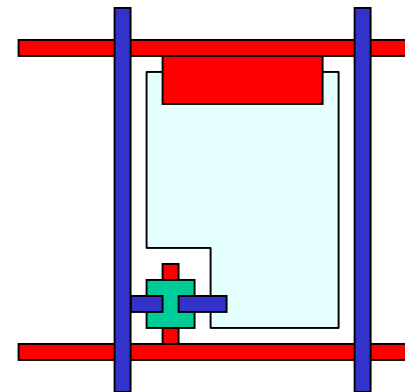
Storage on Common vs. Storage on Gate



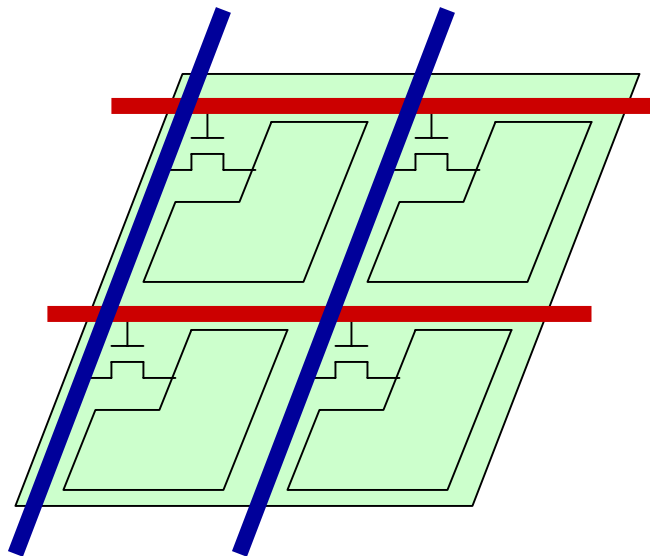
Storage on Common



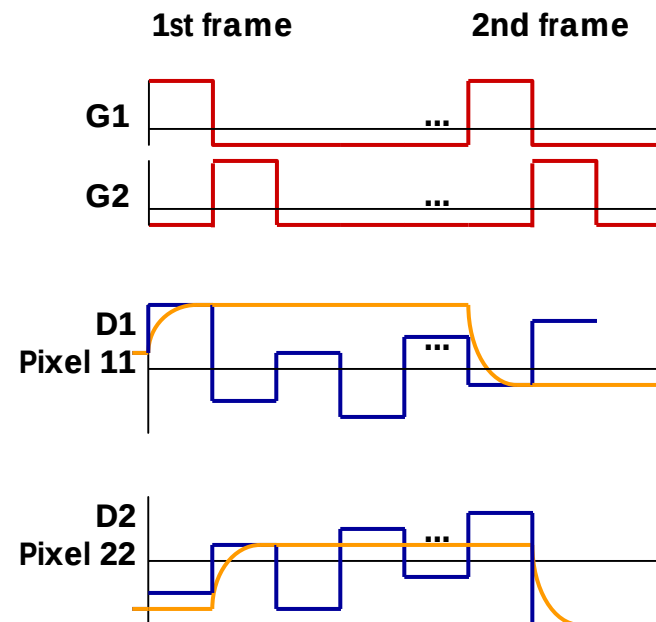
Storage on Gate



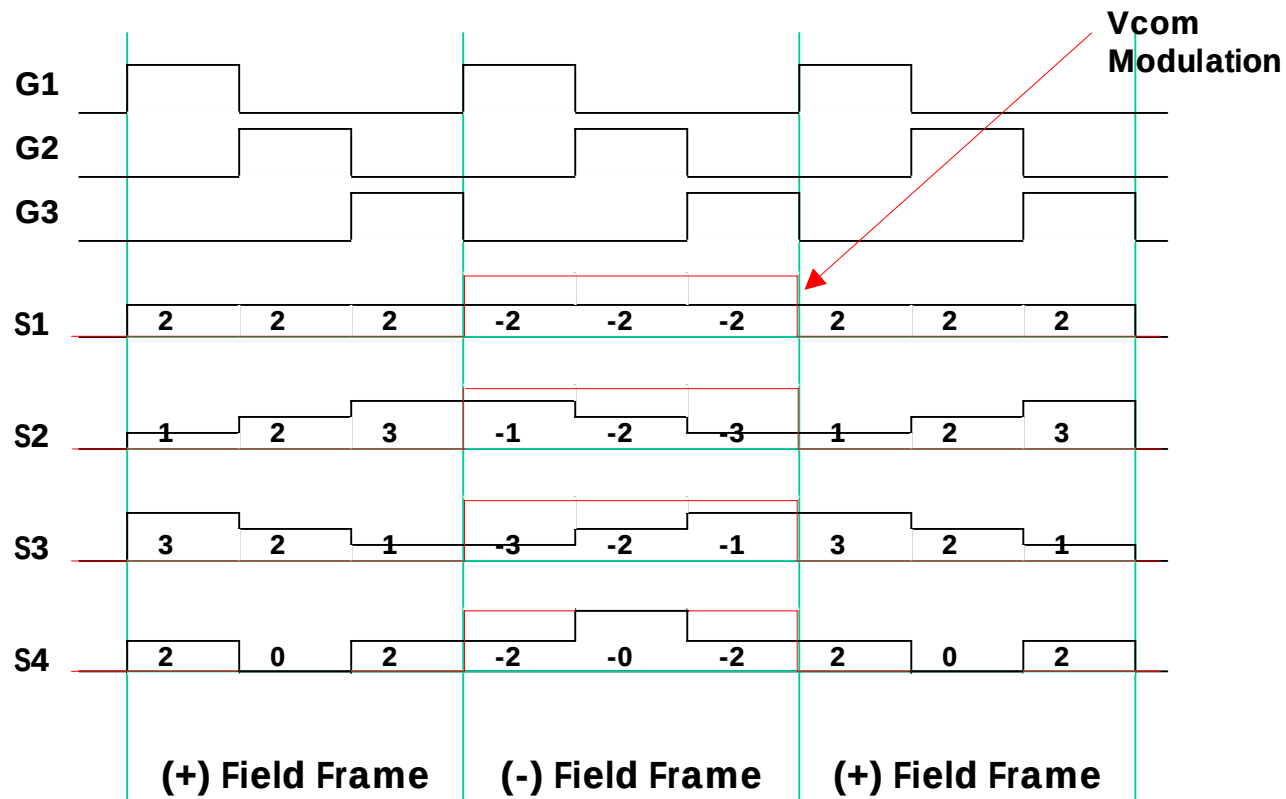
Active Matrix with a TFT as Pixel Switch



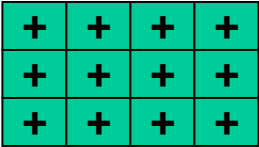
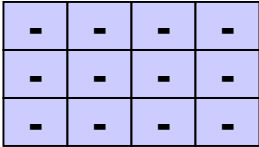
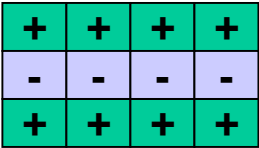
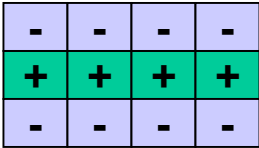
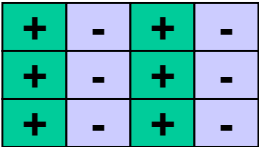
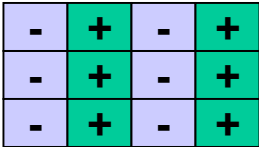
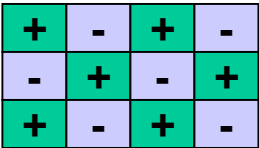
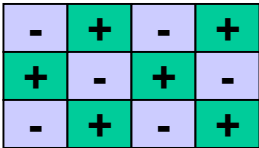
응답 속도 → 12 ms



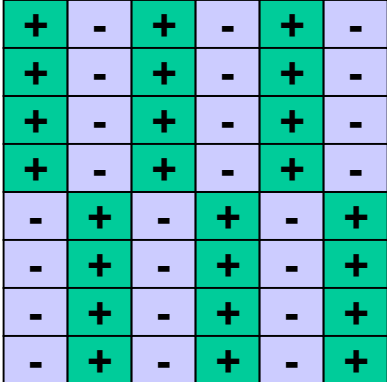
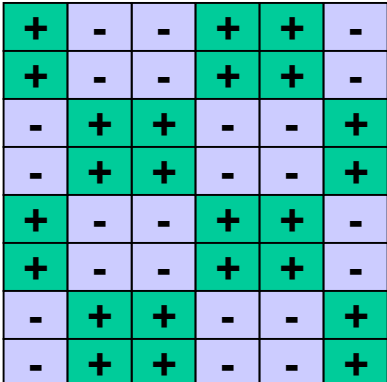
Active Matrix Addressing (Frame Inversion)



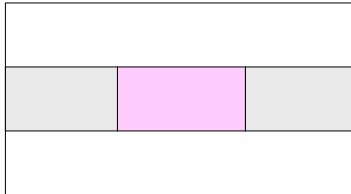
Pixel Inversion Methods

First Frame (+)	Second Frame (-)	Name
		Frame Inversion Low Power Vcom Modulation
		Line Inversion Reduction in Flickers Vcom Modulation
		Column Inversion Low Power Reduction in Flickers
		Dot Inversion No Flickers, No Crosstalks

Some Combinational Inversions

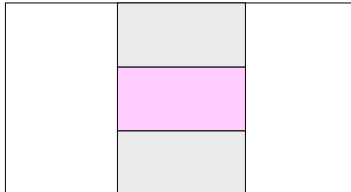



Crosstalk



Horizontal Crosstalk

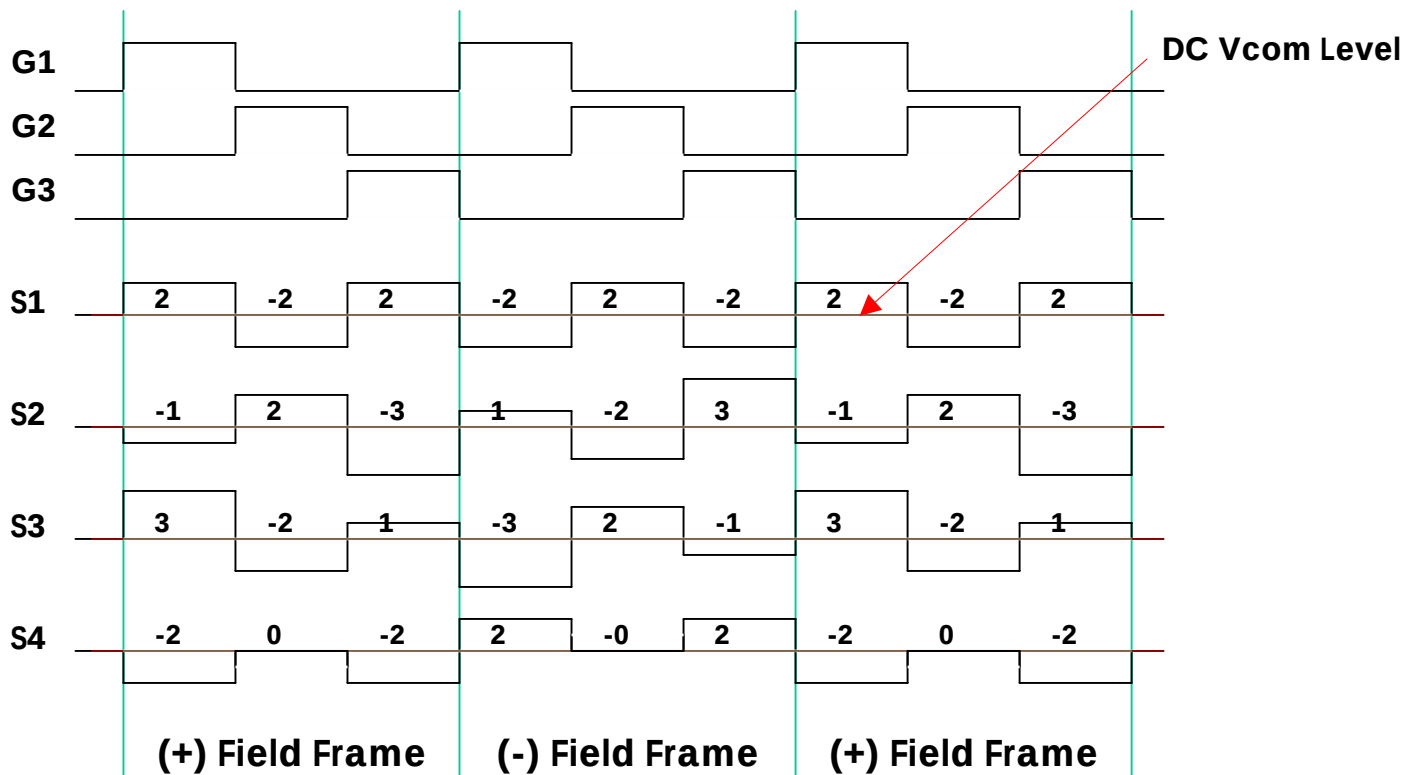
- **Caused by common voltage distortion owing to source-common signal coupling.**
- **Column inversion or dot inversion is effective to reduce it.**



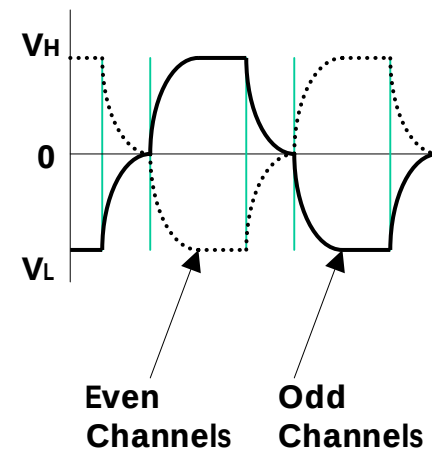
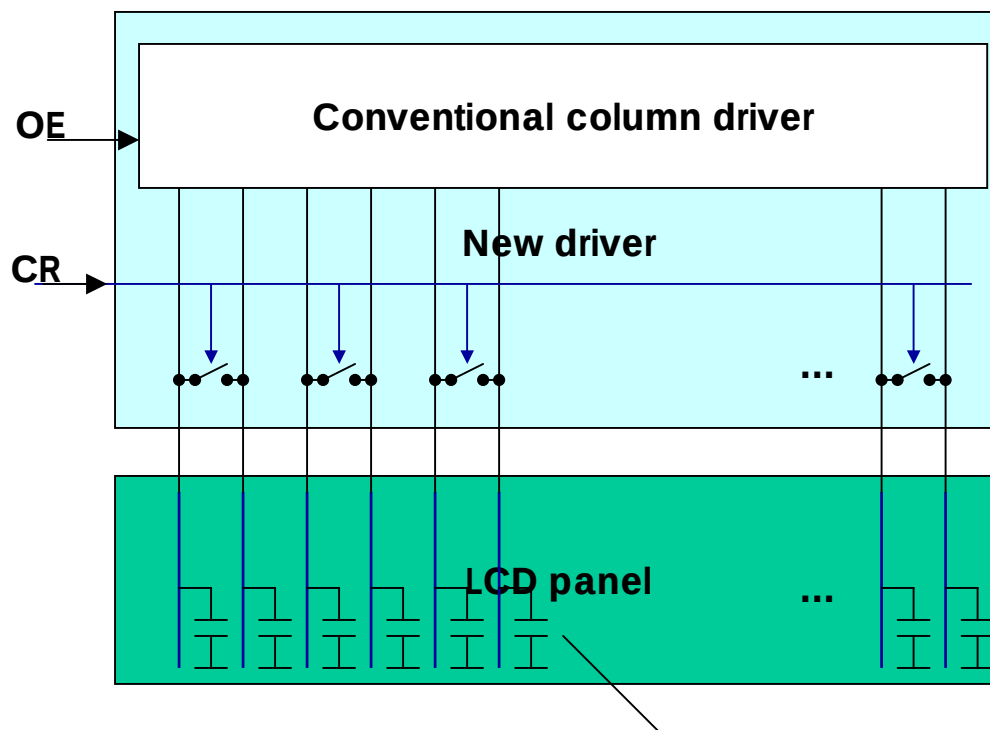
Vertical Crosstalk

- **Caused by source-drain capacitance or leakage current of TFT.**
- **Gate line inversion or dot inversion is effective to reduce it.**

Active Matrix Addressing (Dot Inversion)

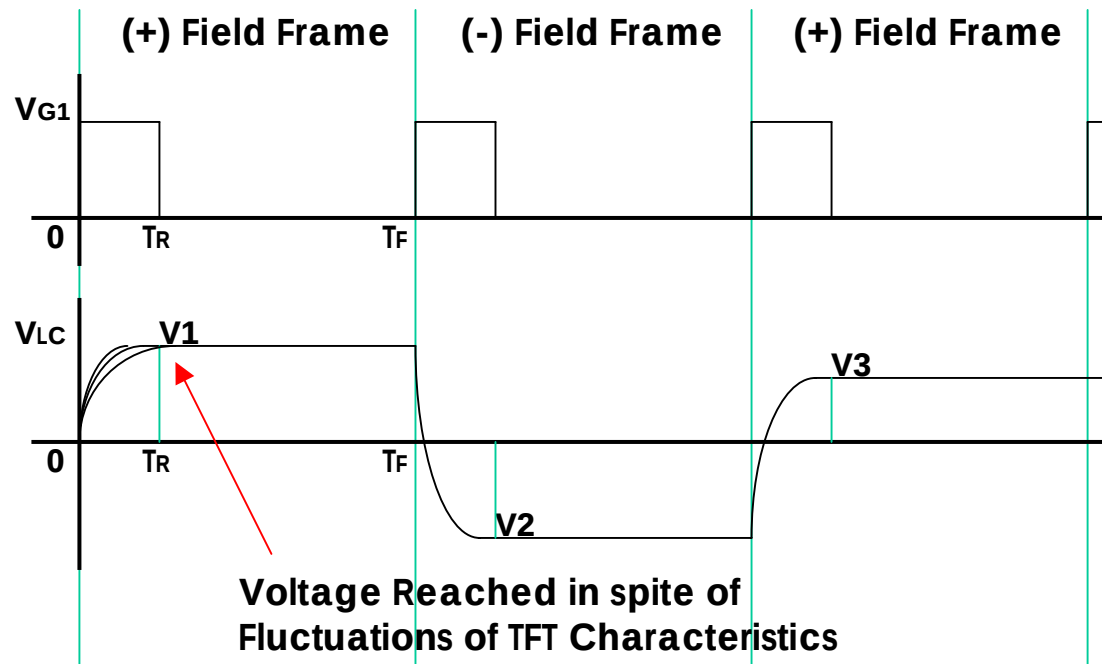


Charge Sharing for Power Recycling



Parasitic capacitances induced from the long column electrode (20~150 pF)

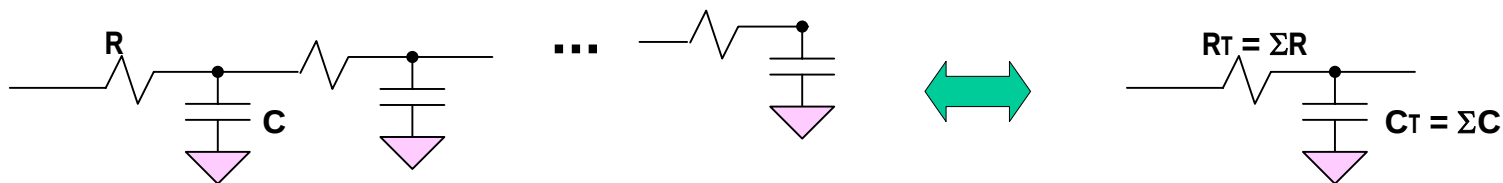
Charging the Storage Capacitor



- $R_{ON} (C_{LC} + C_s) < 0.1 T_R = 0.1 T_F / N$
- $R_{OFF} (C_{LC} + C_s) > 100 T_F$
- $R_{OFF} / R_{ON} = I_{ON} / I_{OFF} > 1000 N$

For 1% Accuracy

The Load Model of the Gate and Data Line



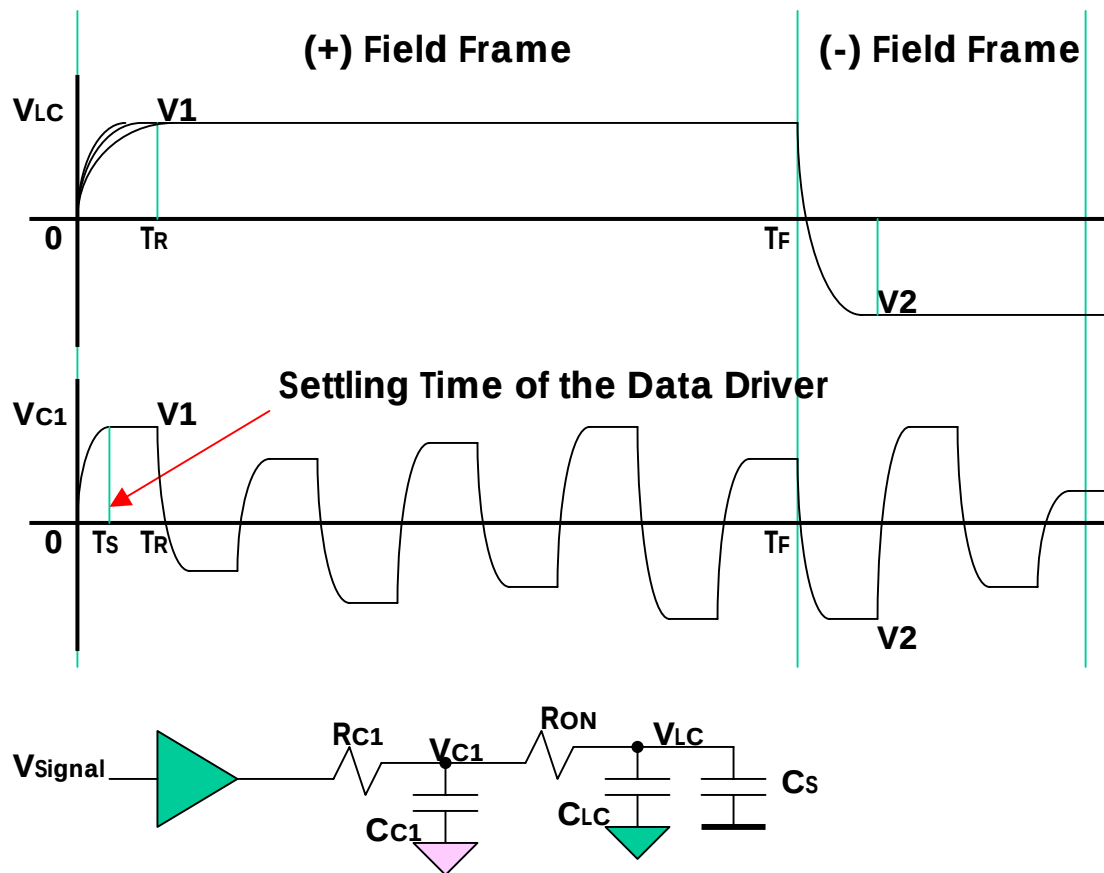
Gate Line Capacitance

- **Cs (Storage Capacitance)**
- **Cg (Gate Capacitance)**
- **Cgc (Gate to Common Electrode)**
- **Cgd (Gate to Data Line Overlap)**

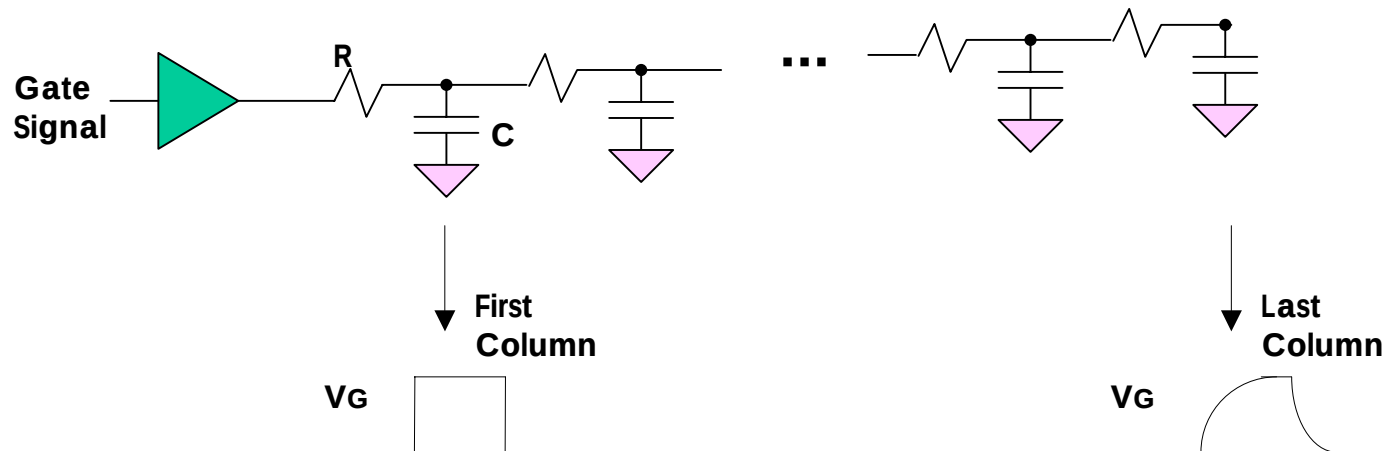
Data Line Capacitance

- **Cgs (Gate-Source Overlap Capacitance)**
- **Cdp (Data Line to Pixel Electrode)**
- **Cdc (Data Line to Common Electrode)**
- **Cgd (Gate to Data Line Overlap)**

Charging of the Column Line Capacitance

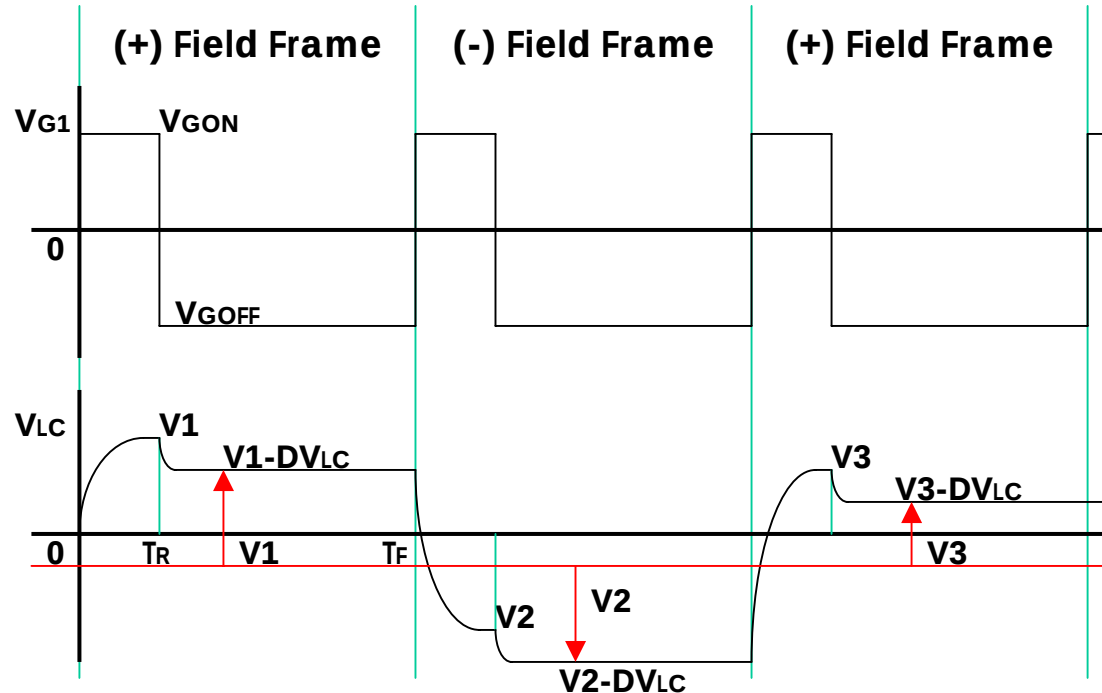


The R-C Delays of Gate Impulses



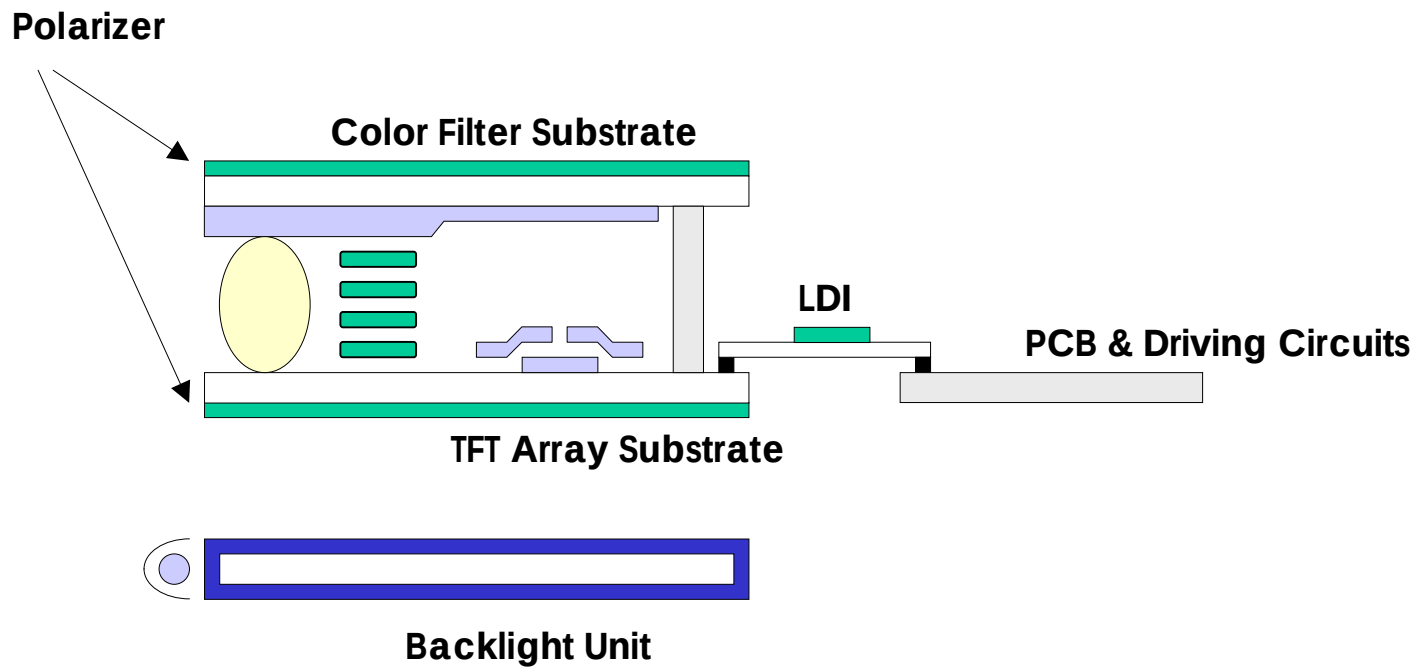
- **Luminance non-uniformity**
- **Flickers**
- **Crosstalks**

The Gate Impulse Effect on the Pixel Voltage

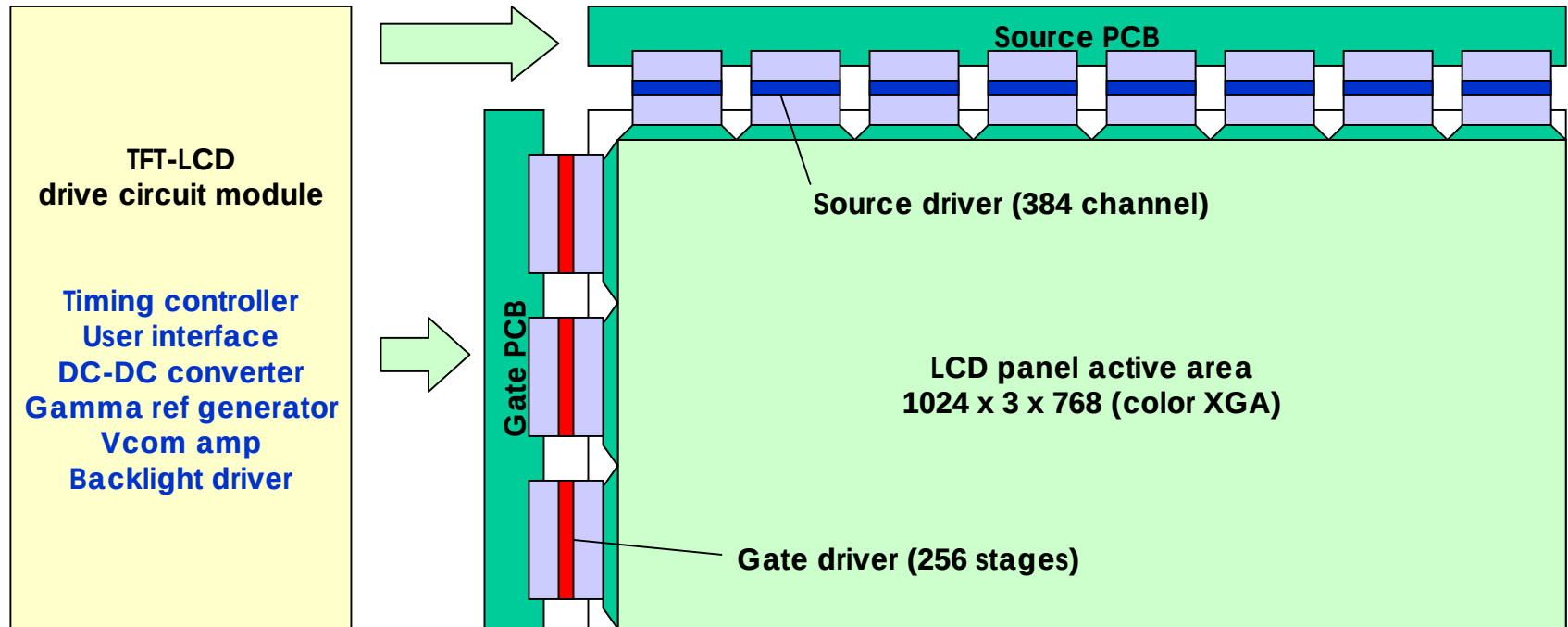


- $DV_{LC} = (V_{GON} - V_{GOFF}) C_{GS} / (C_{GS} + C_{LC} + C_s)$
- A Remedy : $V_{com} = -DV_{LC}$ (C_{GS} , C_{LC} and DV_{LC} depend upon the video voltage.)

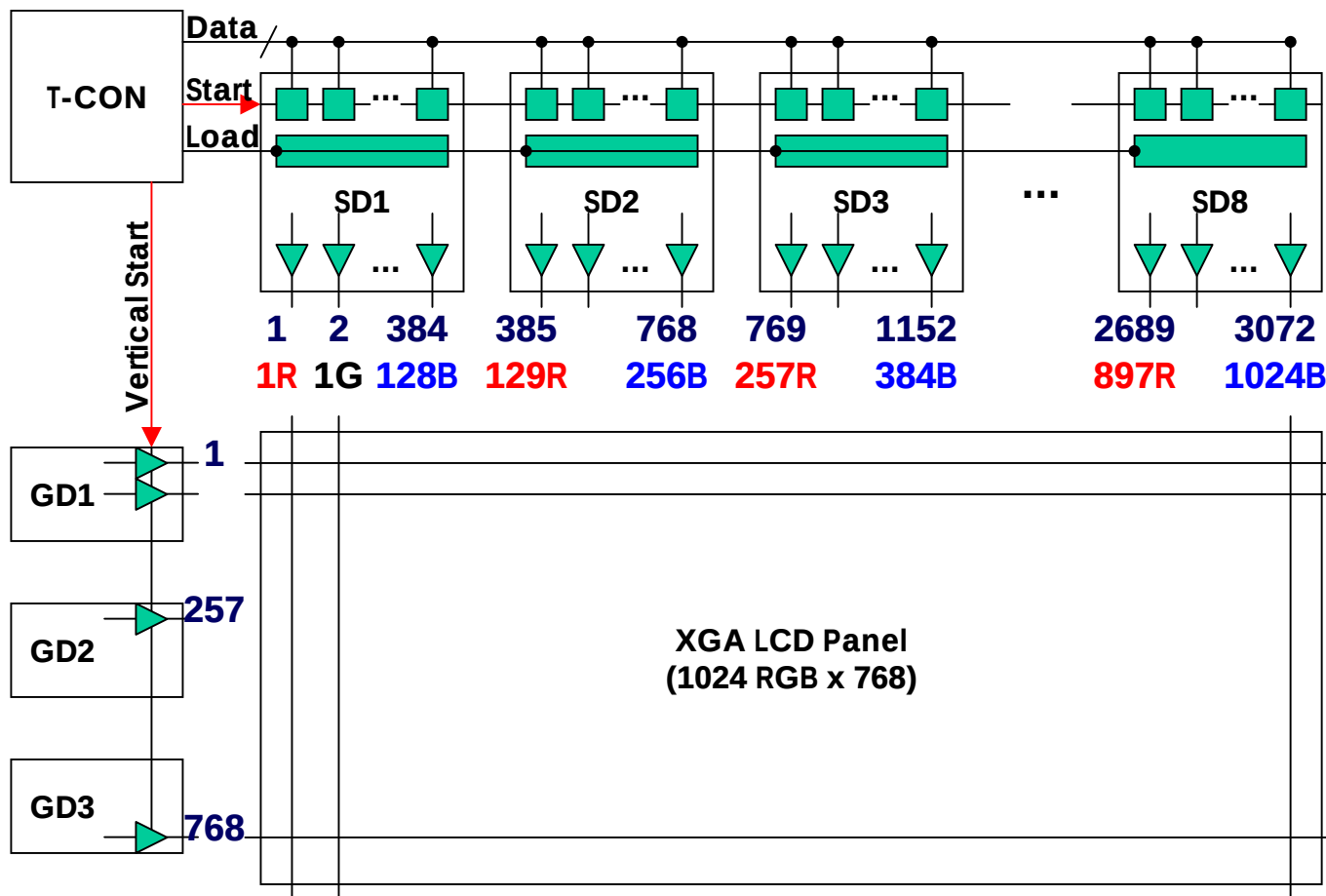
LCD Module Structure



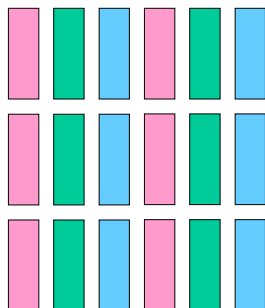
TFT LCD Module



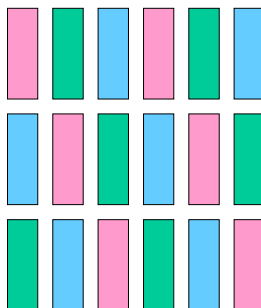
Operation of TFT LCD Module



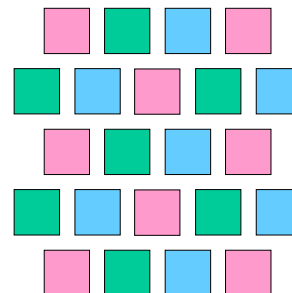
Color Implementation



Stripe
Simple
Poor Color



Mosaic
Intermediate



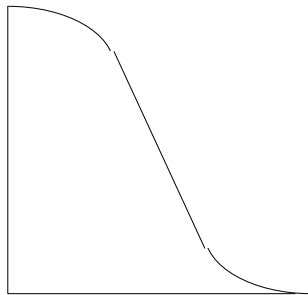
Delta
Complex
Best Color

9 bit color	3-bit RGB	8-level RGB	512 colors
12 bit color	4-bit RGB	16-level RGB	4,096 colors
16-bit color	5-bit R, 6-bit G, 5-bit B		65,536 colors
18-bit color	6-bit RGB	64-level RGB	262,144 colors
24-bit color	8-bit RGB	256-level RGB	16,777,216 colors

Gray Scales

LC Transmittance vs. Voltage

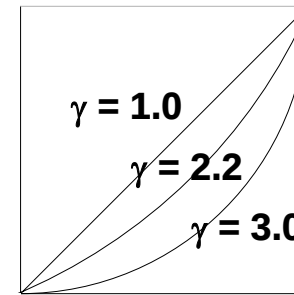
Transmittance



LC Voltage

Light Sensitivity of Human Eye

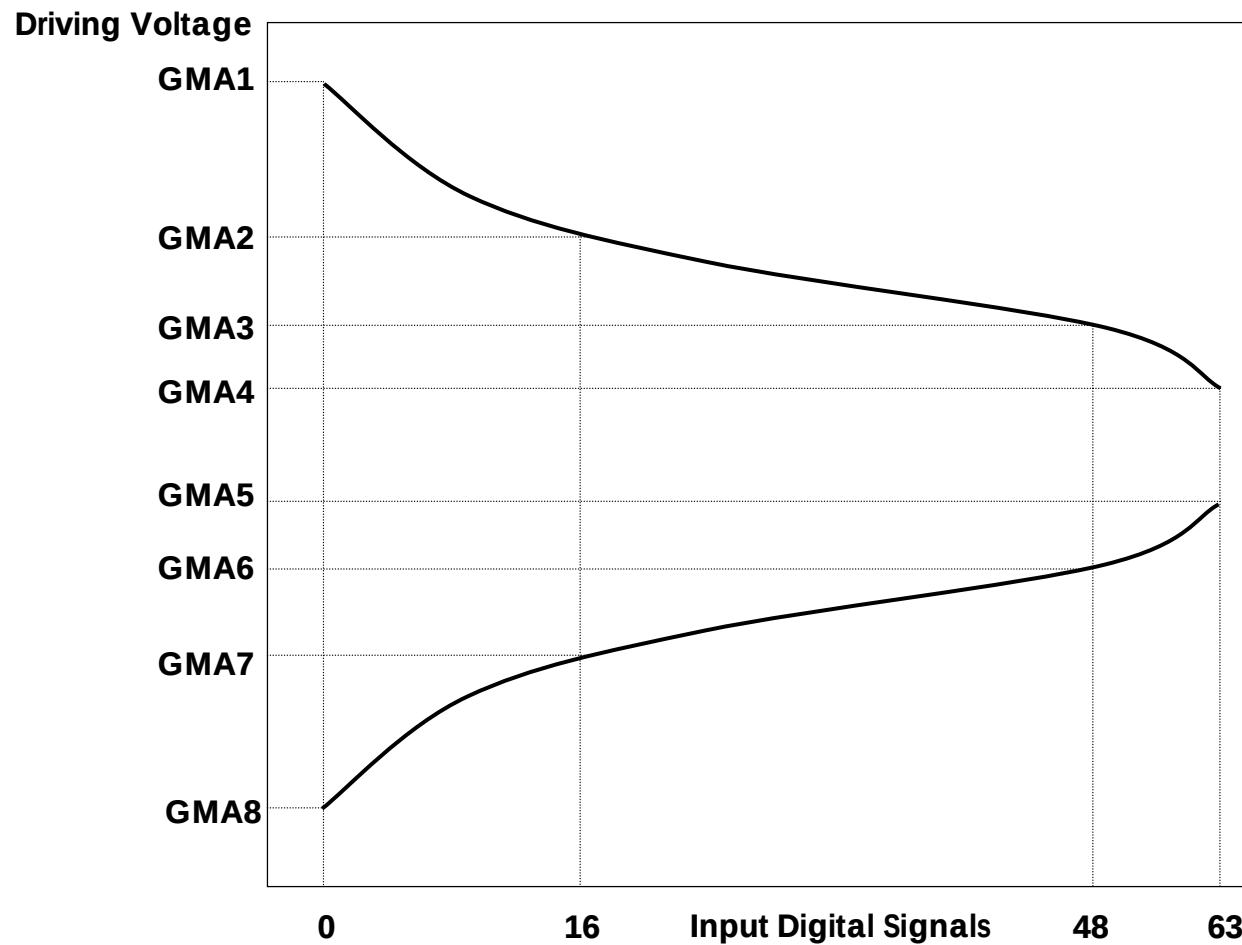
Transmittance



Grayscale

$$T / T_{\max} = (\text{Gray \#} / \text{Max. Gray \#})^\gamma$$

Gamma Adjustment



Increasing the Number of Gray Scales

Dithering Method

30	30	31	30	31	30	31	31	31	31
30	30	30	30	30	31	30	31	31	31
30		30+1/4		30+1/2		30+3/4		31	

No. colors increased by reducing the display resolution.

Frame Rate Control

1st Frame	2nd Frame	
30	30	30
30	31	30+1/2
31	31	31

No. colors increased by increasing the frame rate.
→ High speed addressing is required.

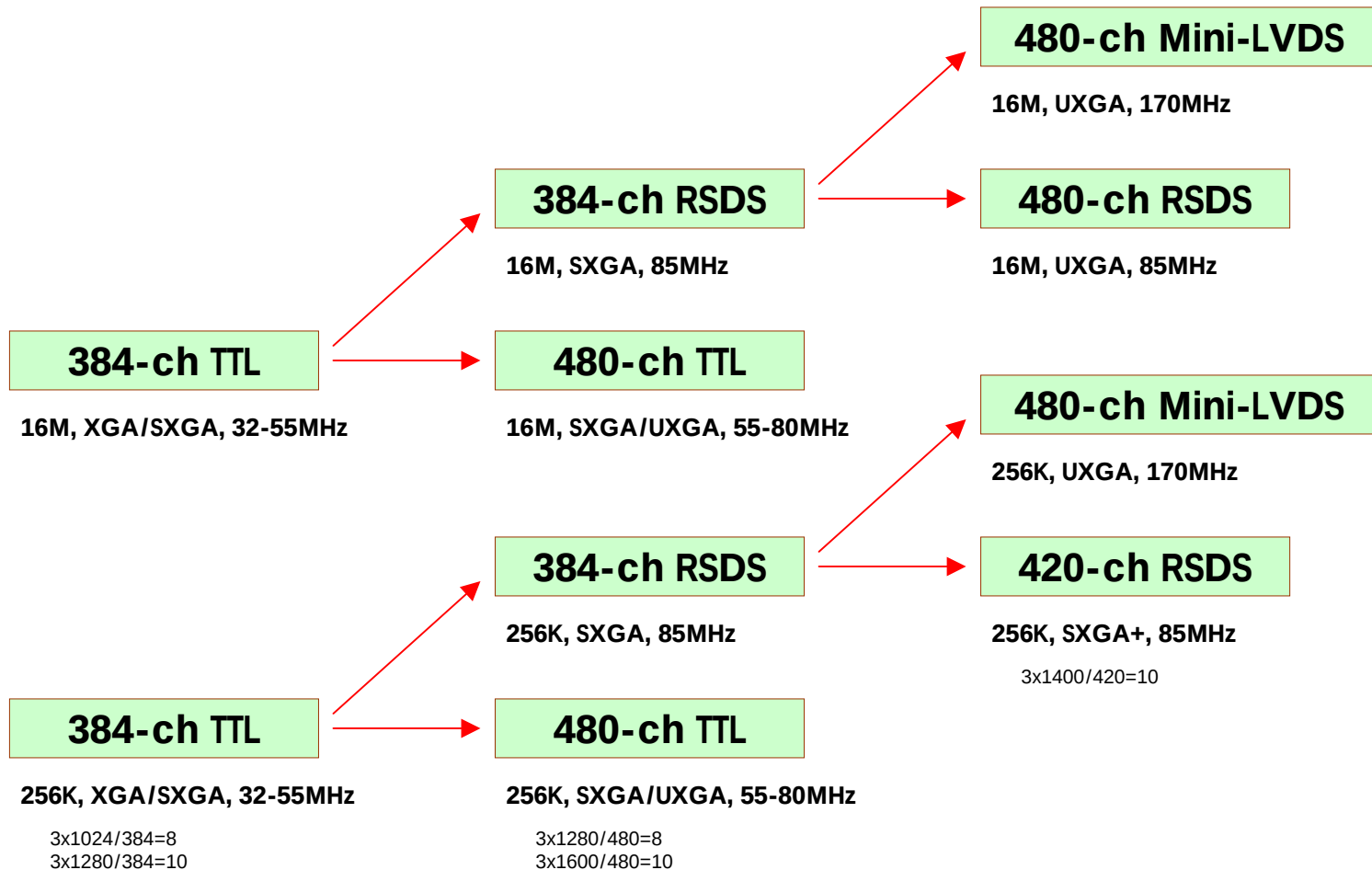
Display Resolution and the System Clock Frequency

60 Hz Frame Rate

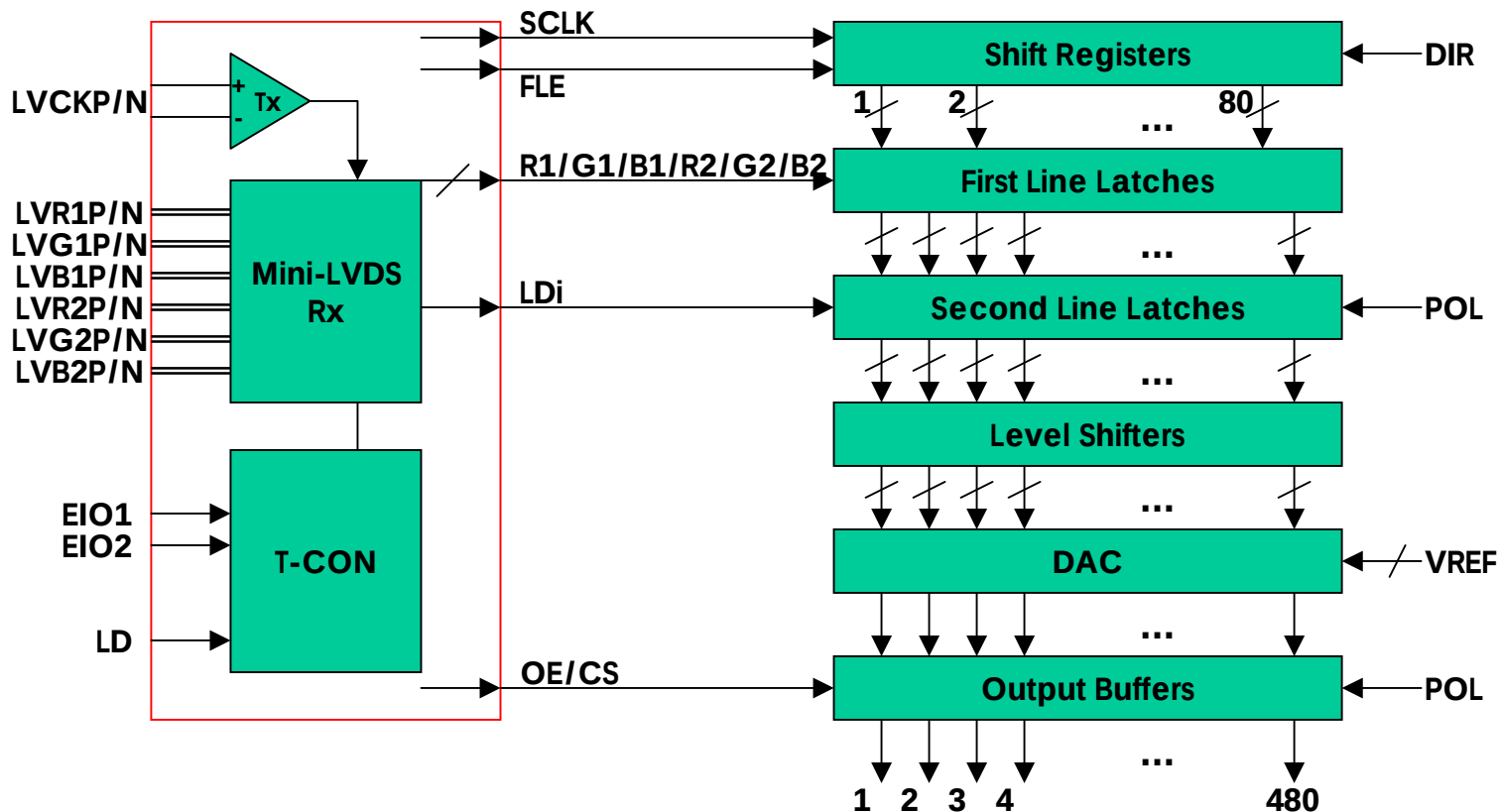
Names	Resolution	Total Screen	V-Sync	System Clock (1 Port)	System Clock (2 Ports)
SVGA	800 x 600	1056 x 628	26.54 us	40 MHz	20 MHz
XGA	1024 x 768	1344 x 806	20.68 us	65 MHz	33 MHz
SXGA	1280 x 1024	1688 x 1066	15.63 us	108 MHz	54 MHz
SXGA+	1400 x 1050	1890 x 1096	15.21 us	124 MHz	62 MHz
UXGA	1600 x 1200	2160 x 1250	13.33 us	162 MHz	81 MHz

- System Clock = (60 Hz) x (# Rows) x (# Columns) x (# Parallel Input Ports)

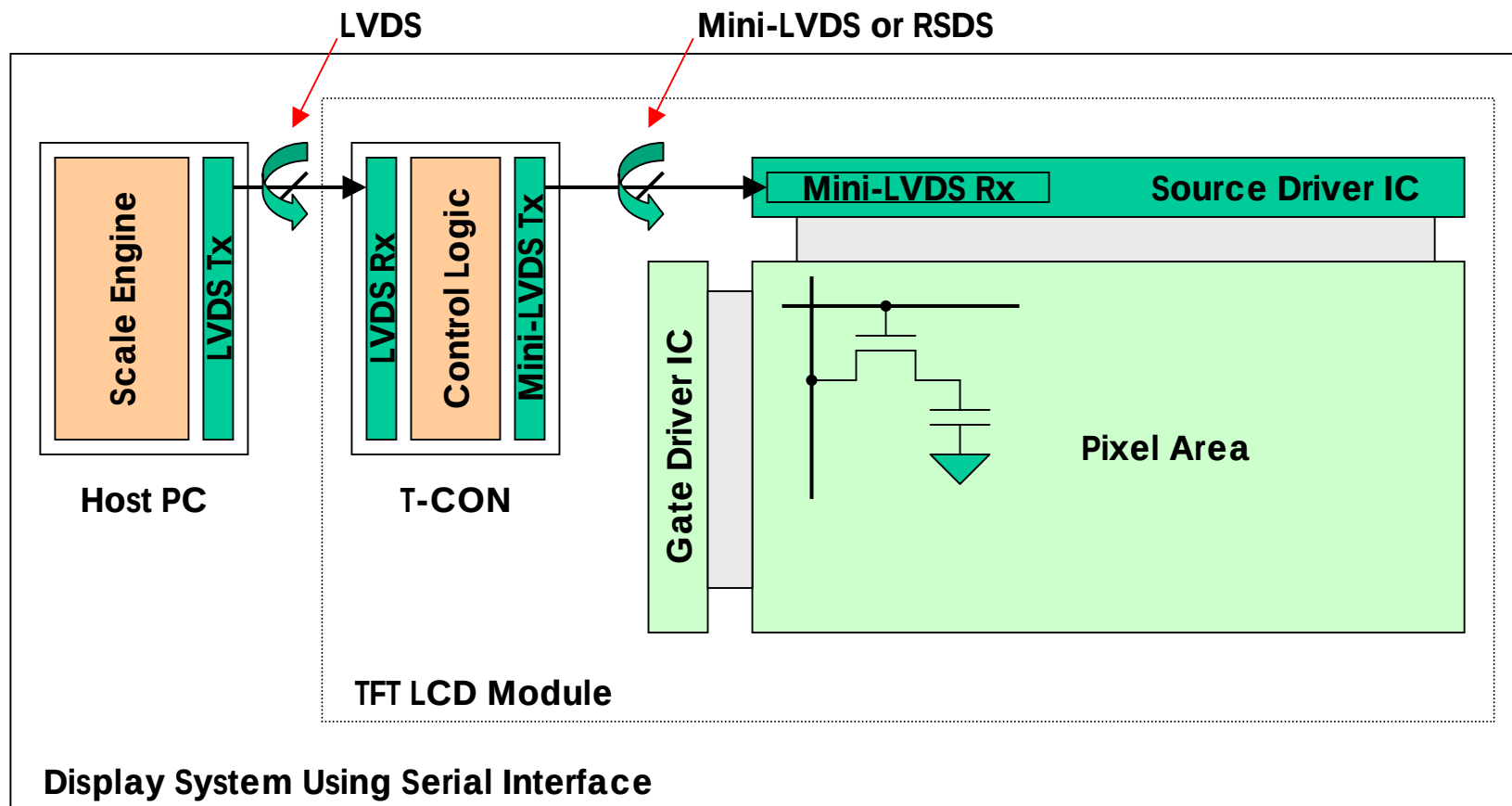
Large Panel Source Driver



Source Driver Block Diagram

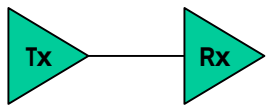
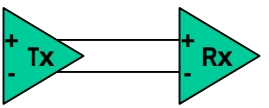
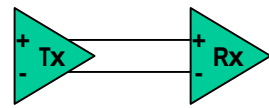
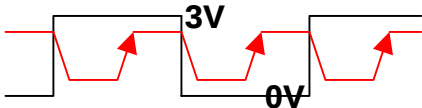
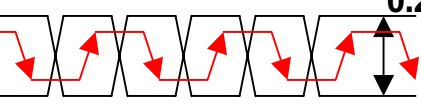
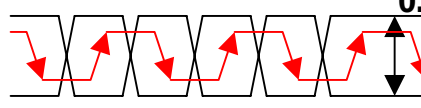
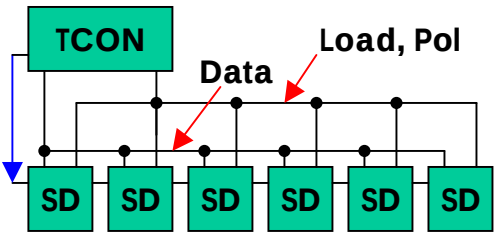
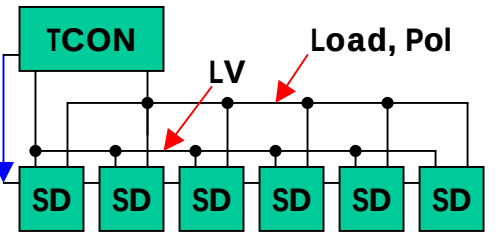
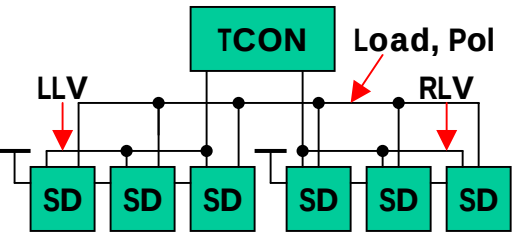


Scopes of Serial Interface



TTL Interface and Serial Interface

XGA 8-Bit RGB (24-Bit) Display System

TTL Interface (CMOS-Level Parallel)	Serial Interface (RSDS)	Serial Interface (Mini-LVDS)
 Single-Ended	 Differential-Signal	 Differential-Signal
 CMOS-Level Square Wave	 Low-Voltage Differential Signal	 Low-Voltage Differential Signal
		
48-bit (dual RGB) @ 33 MHz	24-bit @ 65 MHz (or 48-bit @ 33 MHz)	12-bit @ 65 MHz

Mini-LVDS Data Bit Mapping

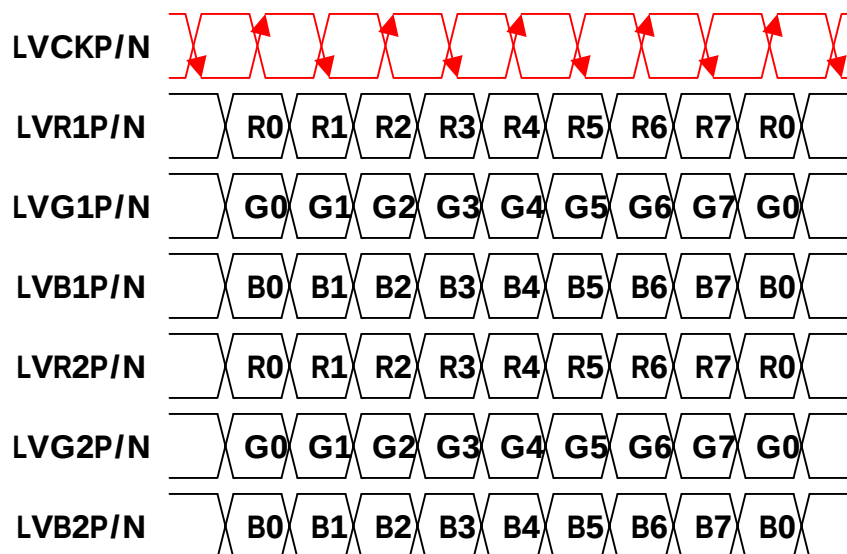
Low-EMI
High bandwidth
Low-cost

Mini-LVDS ← Texas Instrument

RSDS (Reduced-swing differential signaling) ← National Semiconductors

Whisper bus ← Vivid Semiconductor

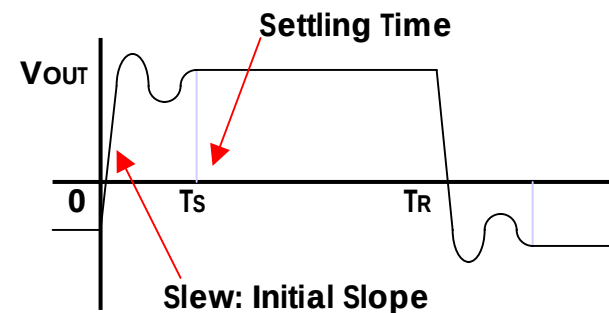
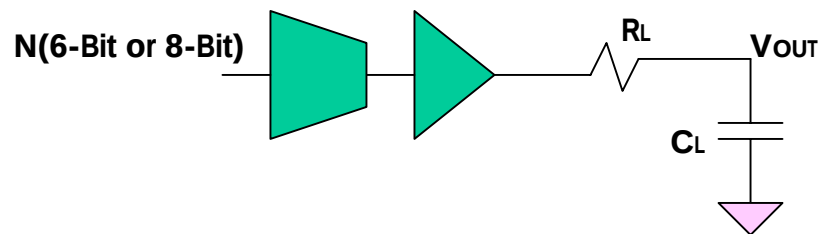
CMADS (current-mode advanced differential signaling) ← NEC



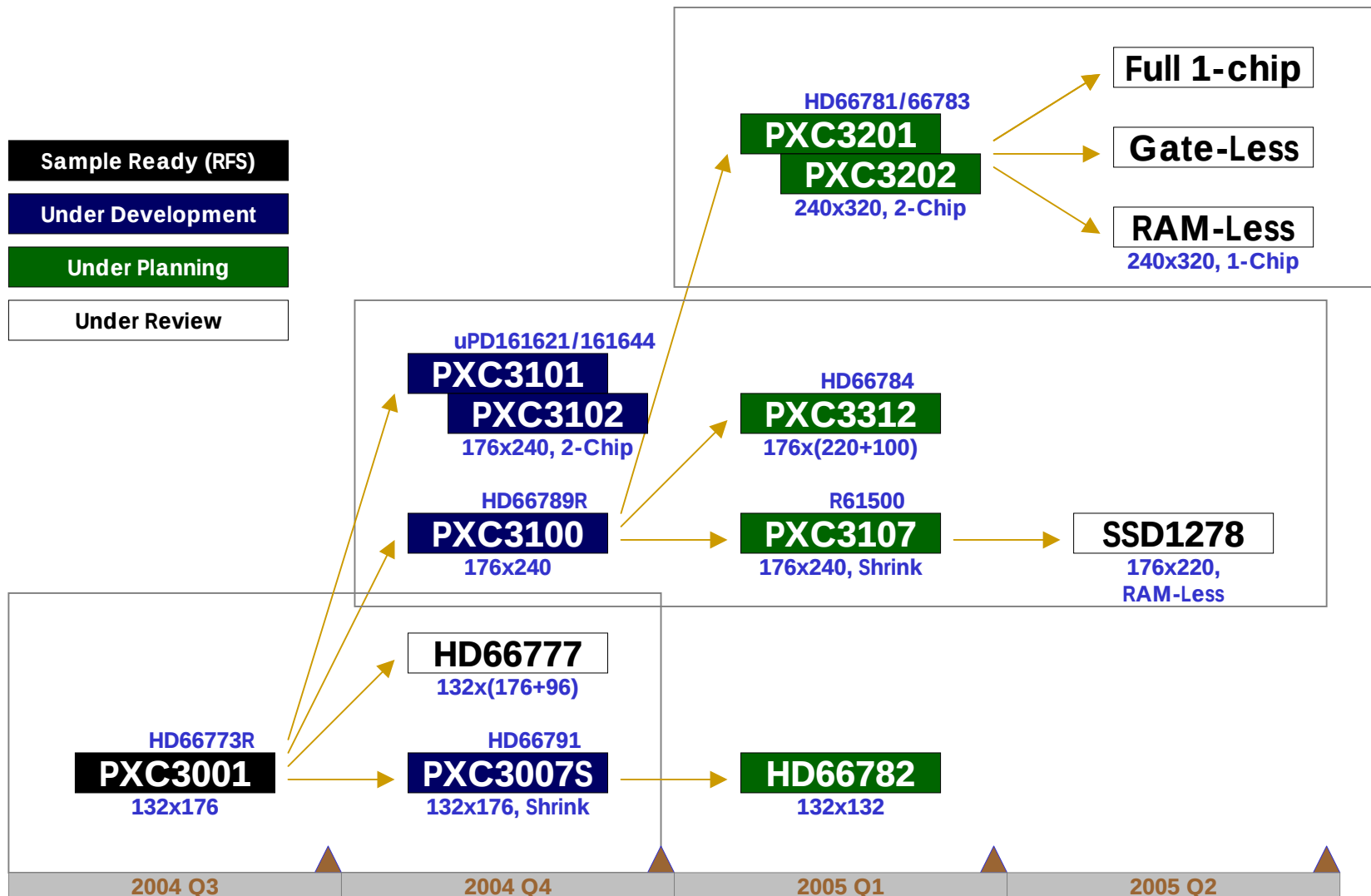
Multiplexing Ratio = 8:1

Source Driver Specifications

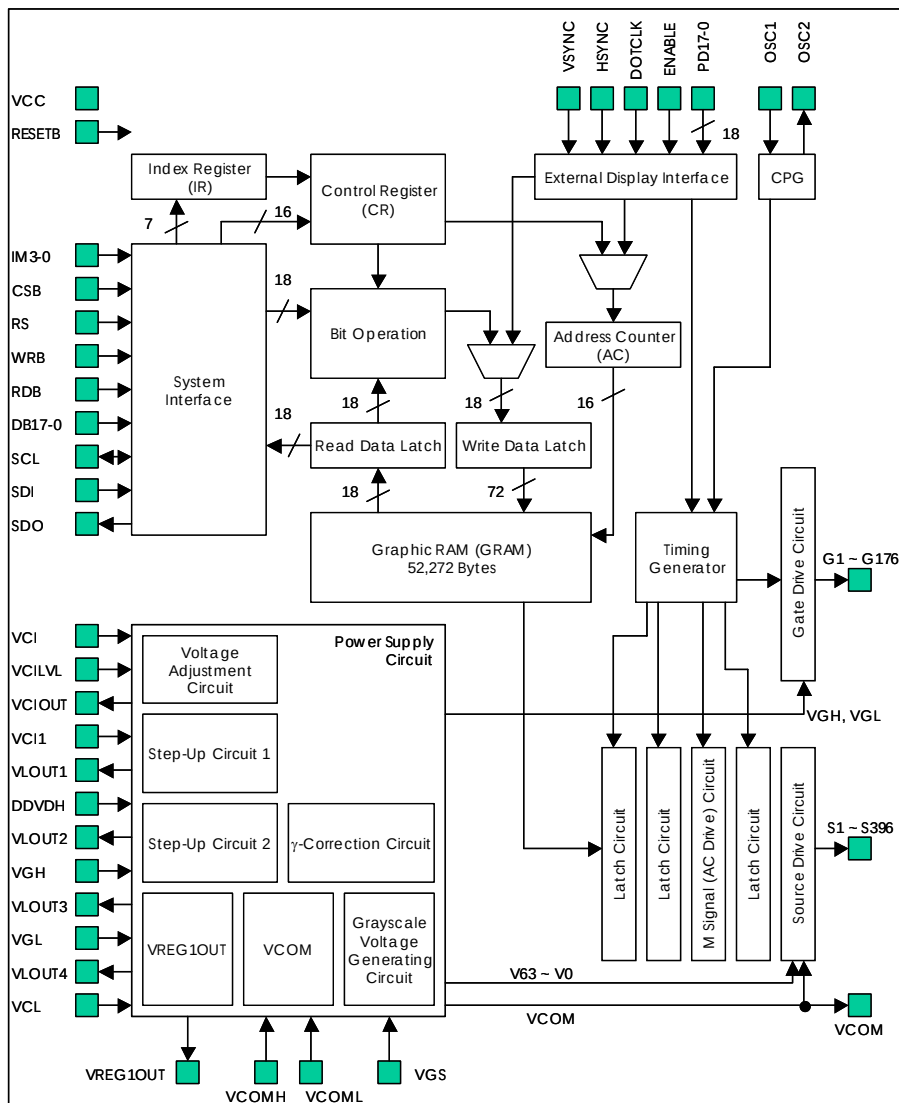
- LCD Operation Voltage
- # of Input Ports (1-Port RGB or 2-Port RGB)
- Maximum Clock Frequency
- Slew and Settling Time at Typical Load Conditions
- Output Uniformity between Channels and between IC's
 - $V_{DE} = (V_{OUTP}(N) - V_{OUTN}(N)) / 2 - V_{TARGET}(N)$
 - $V_{CME} = (V_{OUTP}(N) + V_{OUTN}(N)) / 2 - V_{MIDDLE}$
- Gamma Characteristics



Mobile TFT LCD Driver IC's: Pixelchips



PXC3007S Complexity



One-chip driver for 262,144 color, 132 x 176 TFT-LCD

Full 262,144 colors

18/16/9/8 bit parallel interface (i80 & M68) and SPI

18/16/6 bit RGB interface and VSYNC interface

Internal graphic RAM : $18 \times 132 \times 176 = 418,176$ bits

Window RAM access

Partial display screen and vertical display scroll

Power save mode : standby mode, sleep mode

N raster row inversion drive

Internal oscillation circuit

Supports the storage on common structure only.

Internal power supply circuit: upto 8x step up

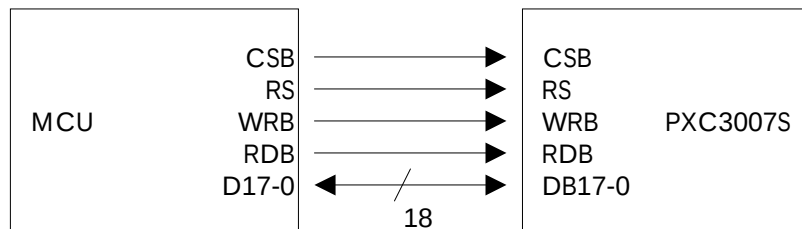
VCI = 2.5V ~ 3.0V

VCC = 2.4V ~ 3.0V

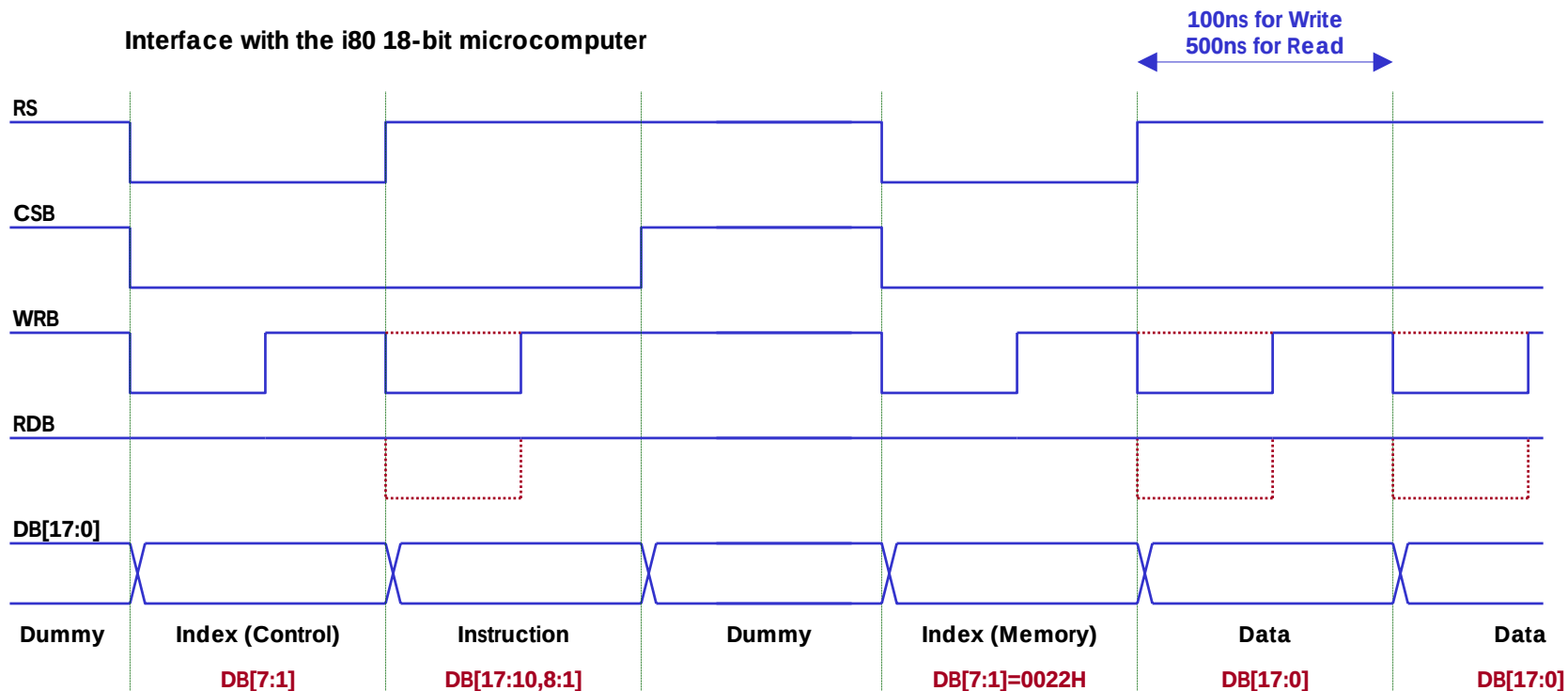
IOVCC = 1.65V ~ 3.0V

AVDD = 4.5V ~ 5.5V

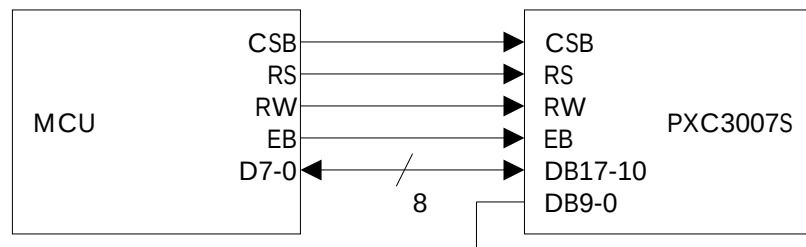
i80 MCU Interface



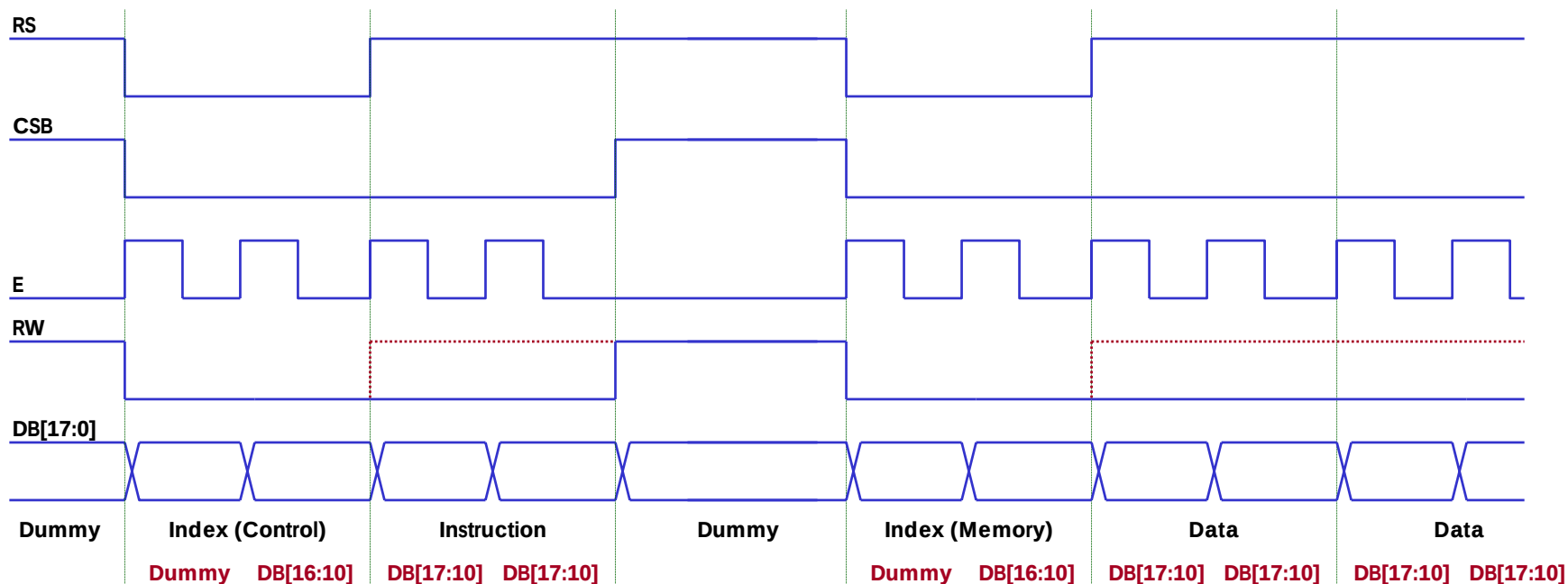
Interface with the i80 18-bit microcomputer



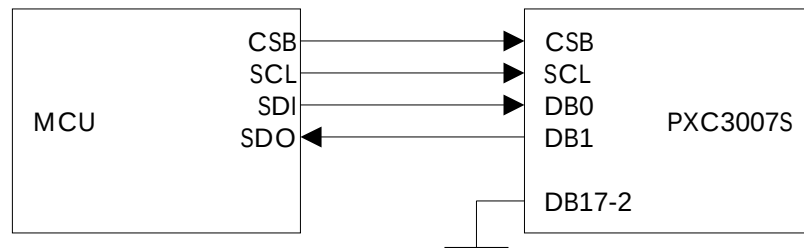
M68 MCU Interface



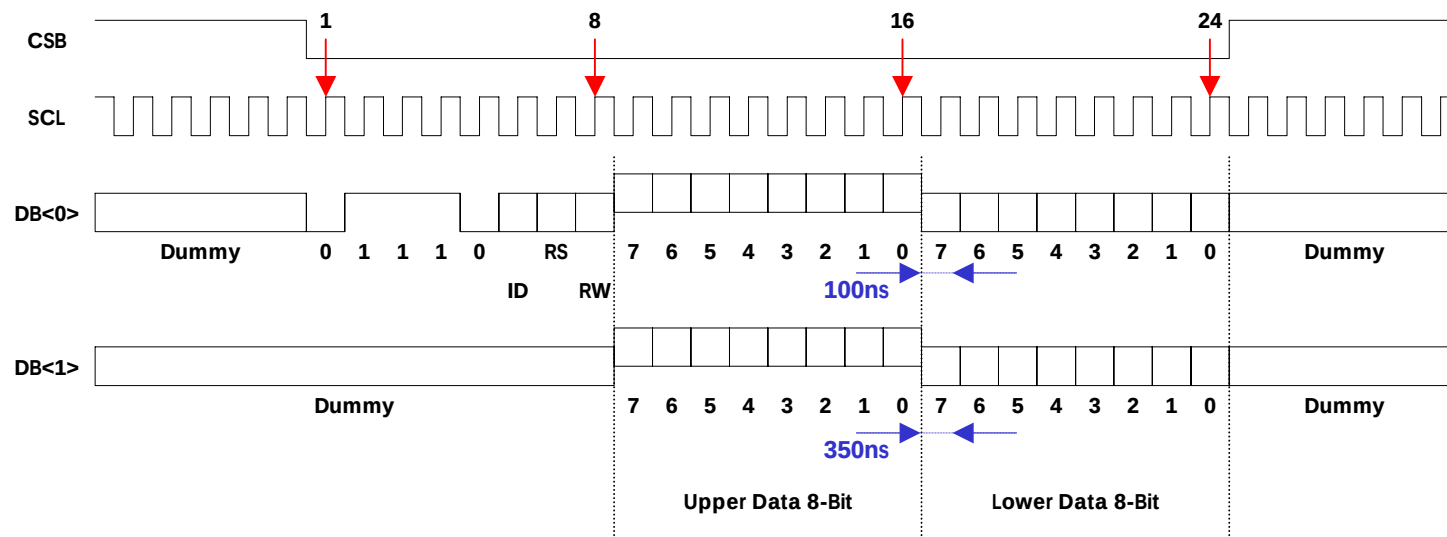
Interface with the M68 8-bit microcomputer (TRI=0)



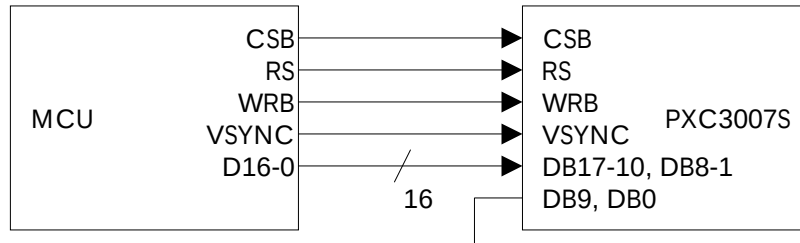
Serial Peripheral Interface



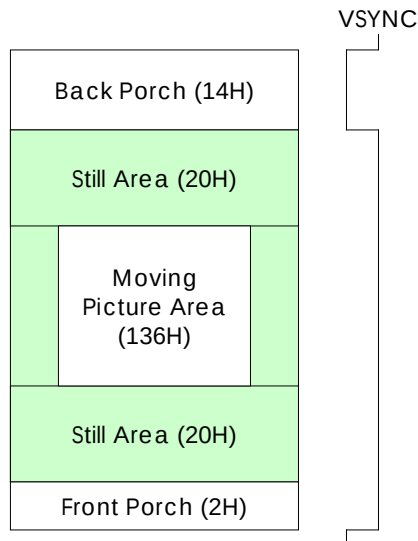
Interface with the Serial Peripheral Interface (TRI=0)



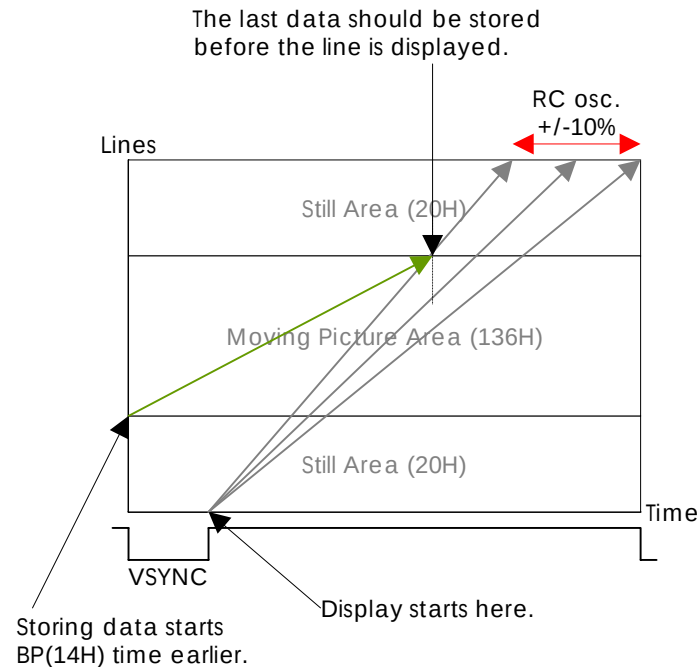
VSYNC Interface



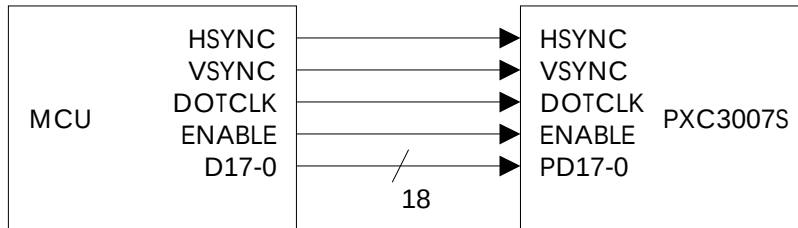
**Data Input Triggered by VSYNC Signal and
Tries to Store the Frame Data Faster than the Display Scan**



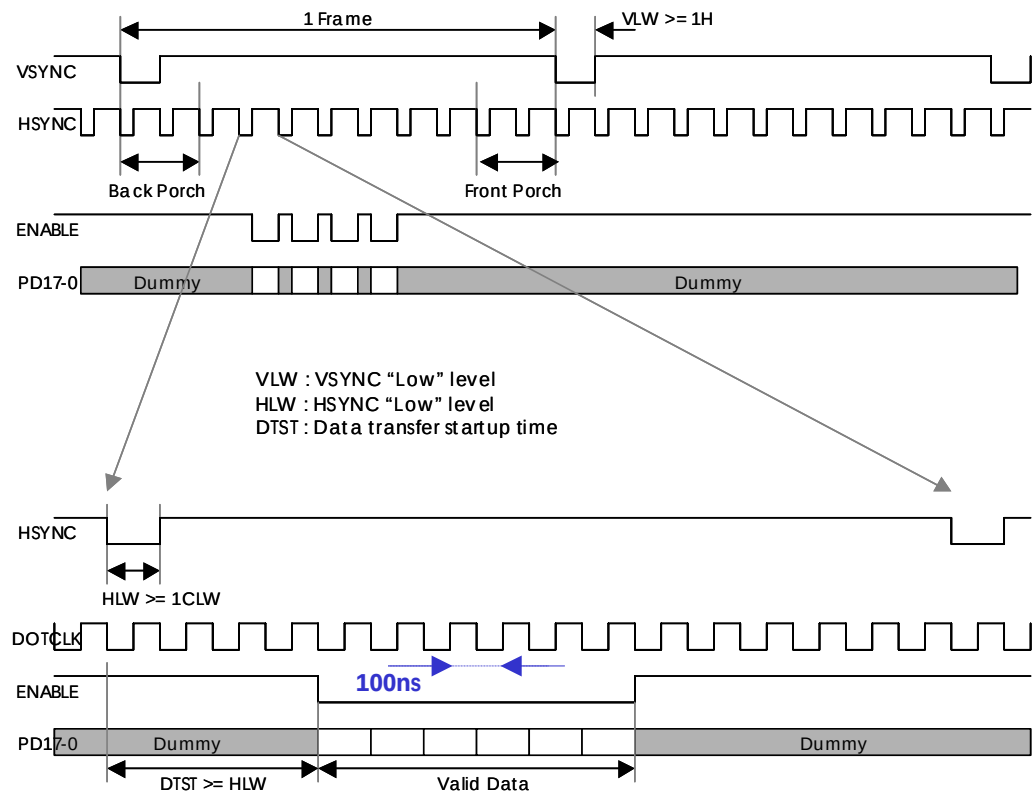
Window Address Function



RGB Interface



18-Bit RGB Interface

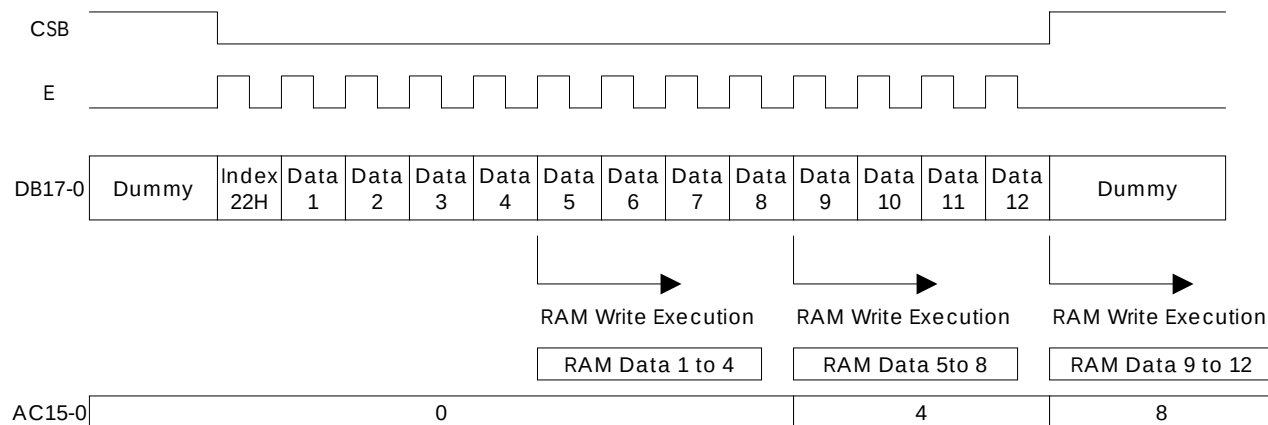


Data $\leftarrow$ RGB Interface
Command $\leftarrow$ Serial Interface

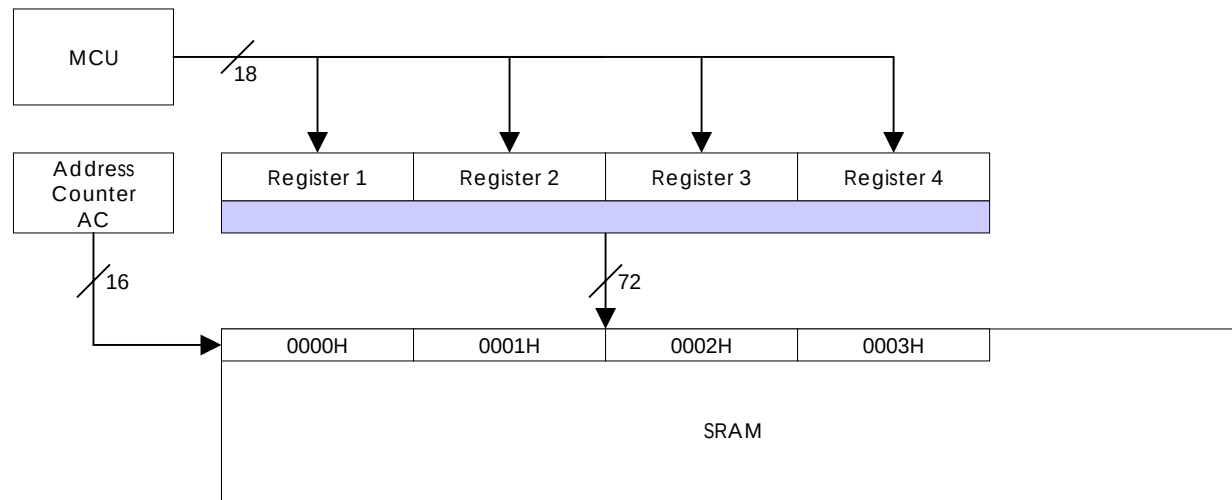
or

Data $\leftarrow$ RGB Interface
Command $\leftarrow$ System Interface
and sharing the DB/PD bus.

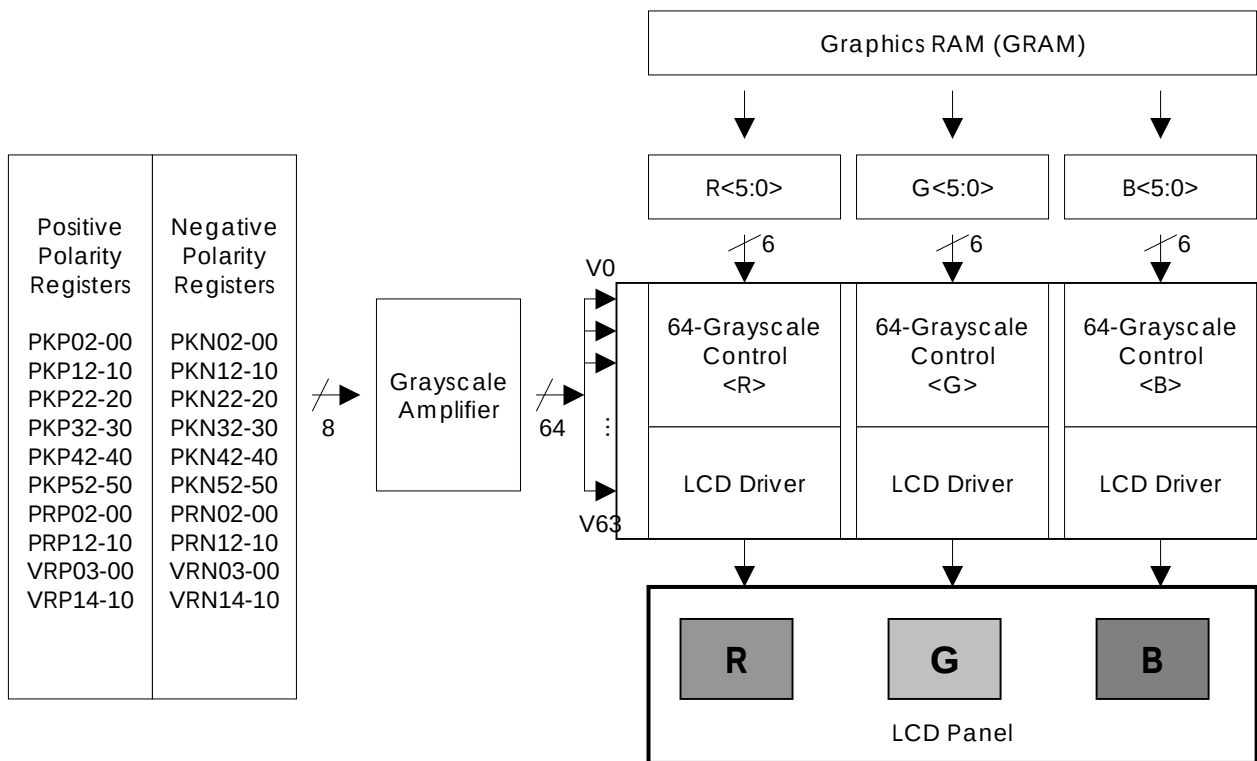
High Speed Write Mode



The speed of memory access is the bottleneck of the data writing.

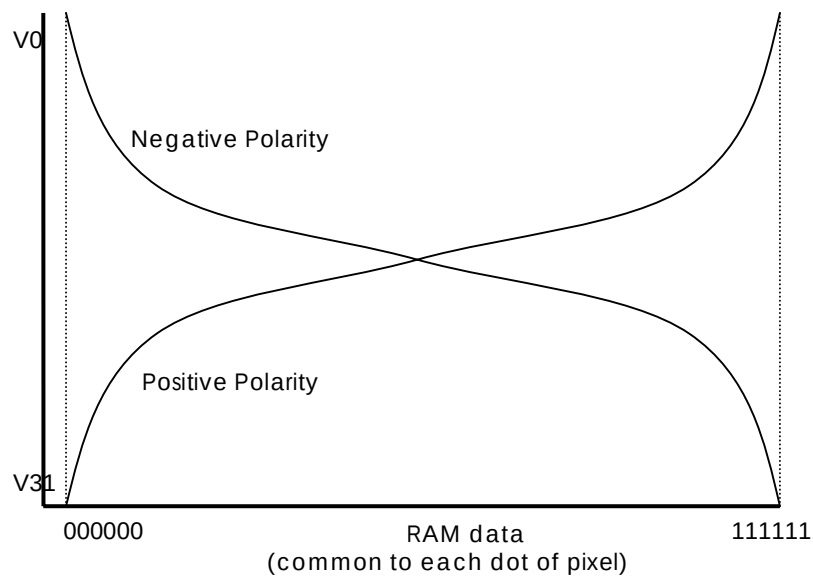


Gray Scale Control

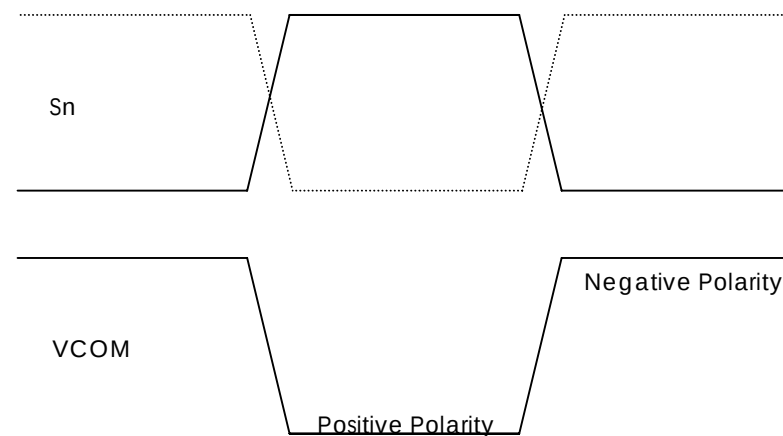


64 Grayscale Amplifiers Rather than 396 Channel Amplifiers.

Source Output and VCOM



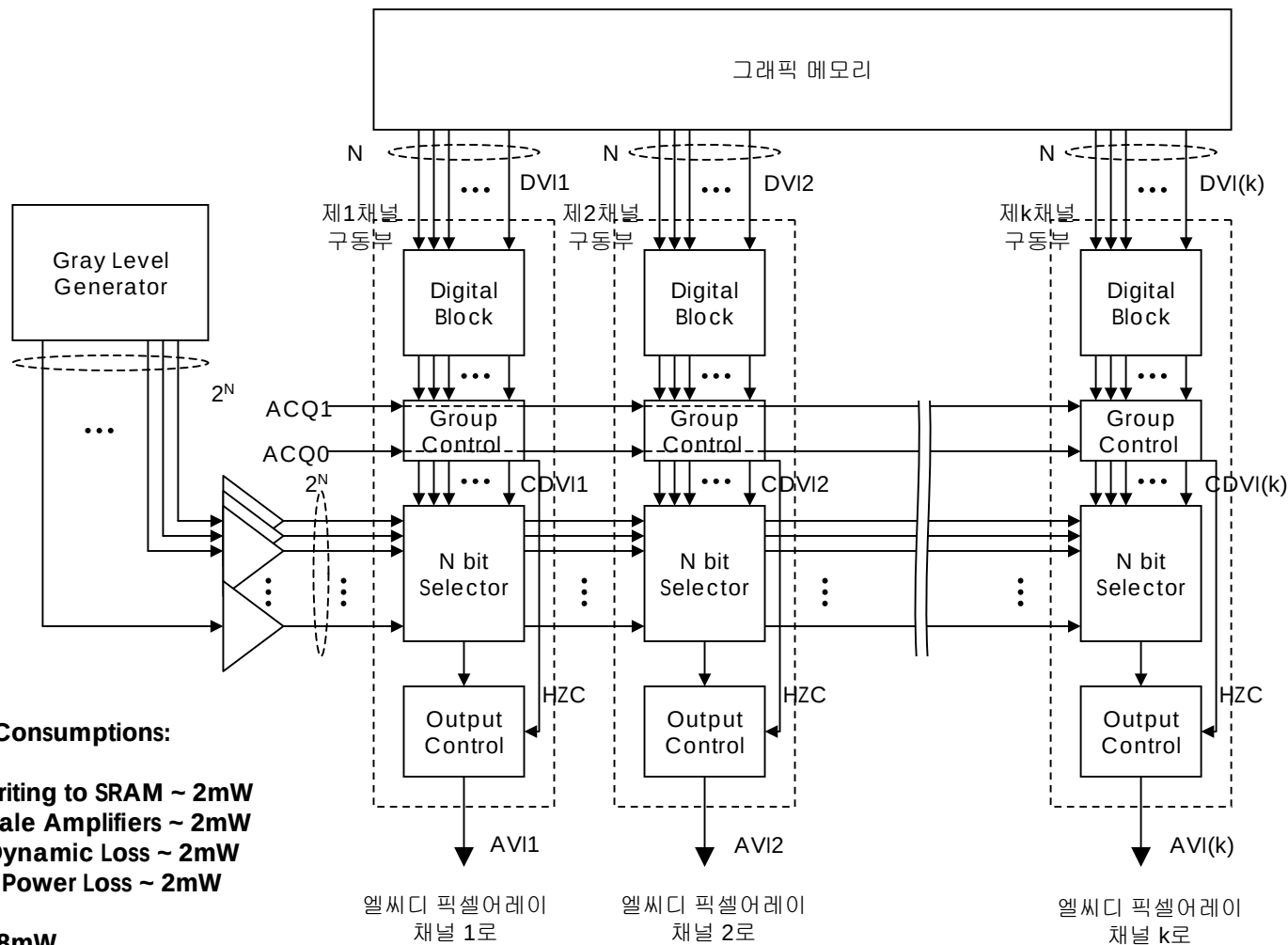
RAM data and the output voltage



Source output and VCOM

Positive polarity gamma and negative polarity gamma can be non-symmetric.

Low Power Grayscale Amp Architecture

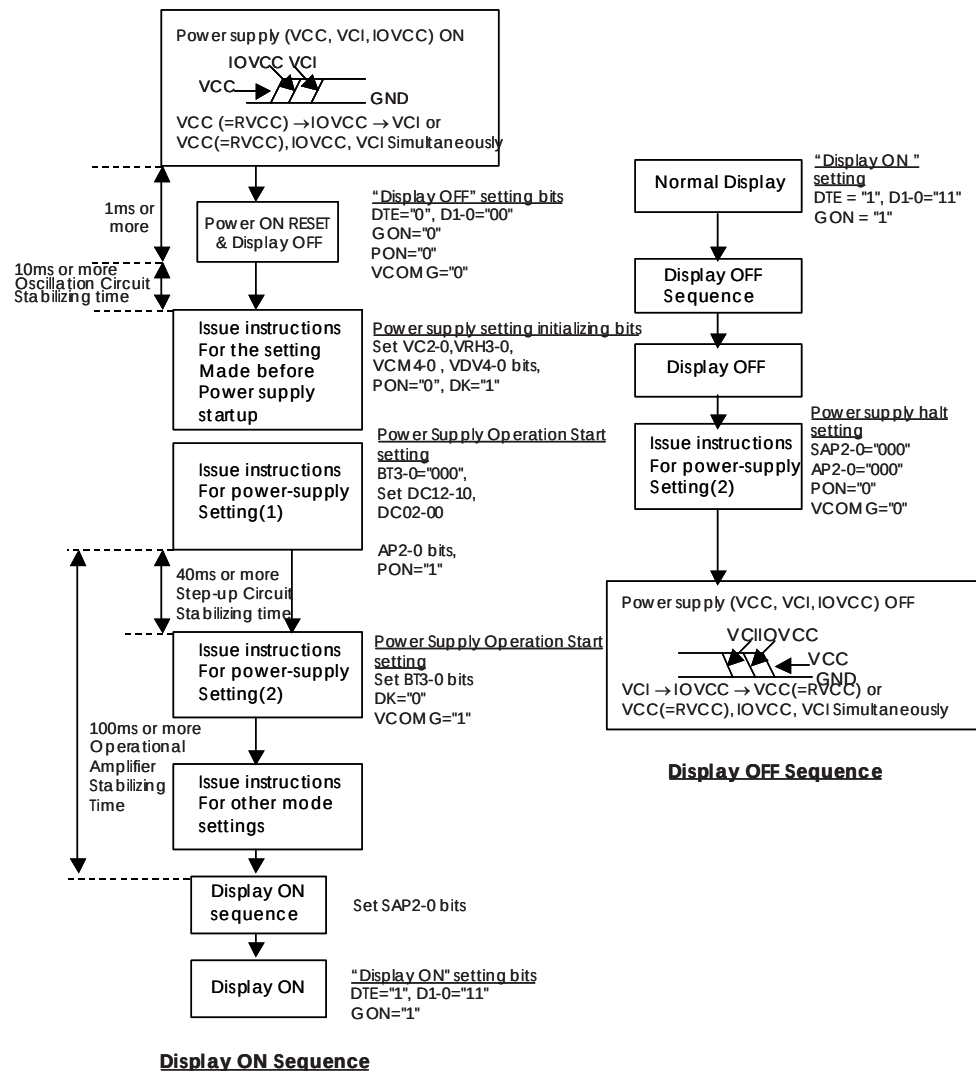


Power Consumptions:

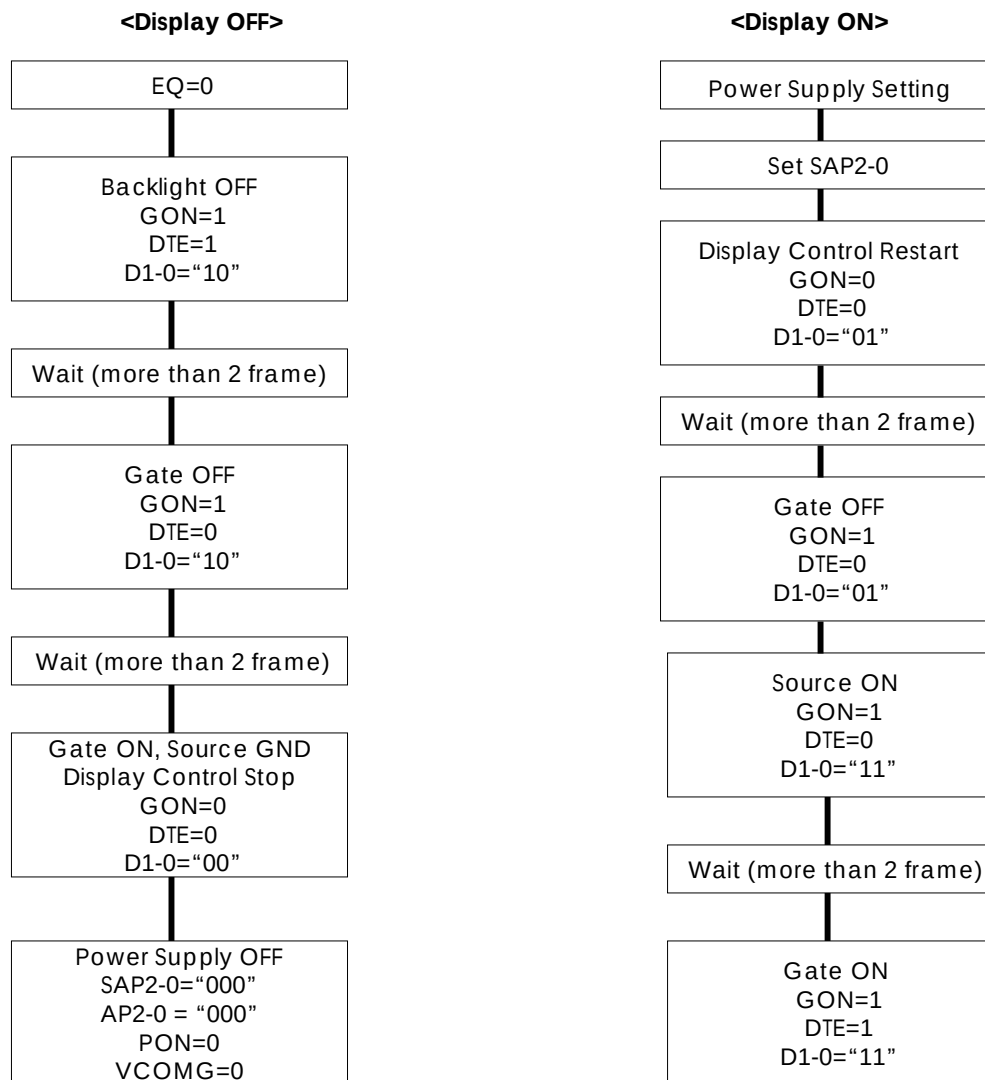
Data Writing to SRAM ~ 2mW
 Grayscale Amplifiers ~ 2mW
 Panel Dynamic Loss ~ 2mW
 DC-DC Power Loss ~ 2mW

Total ~ 8mW
 Excluding the Backlight > 100mW

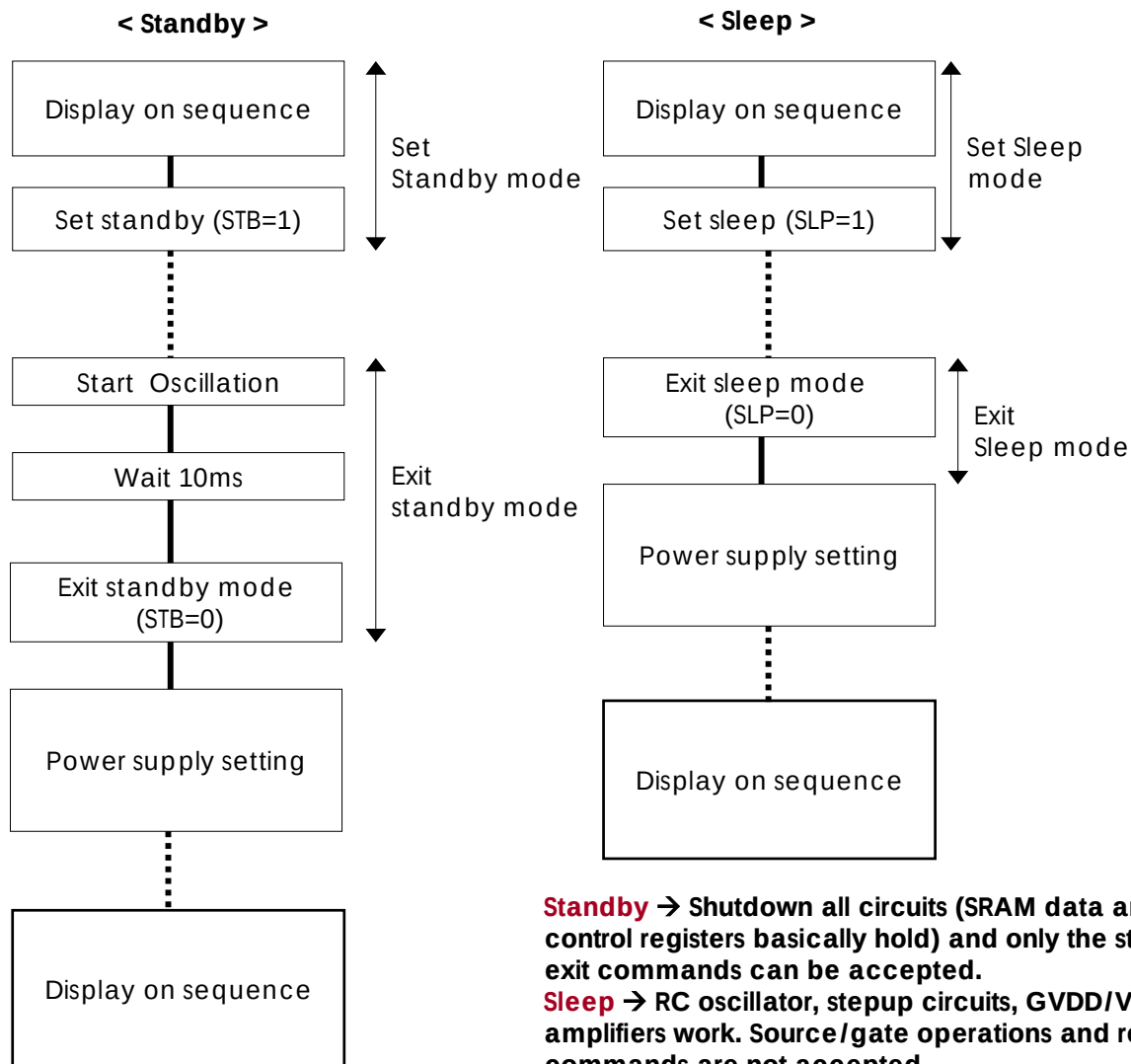
Power ON/OFF Sequence



Display ON/OFF Sequence



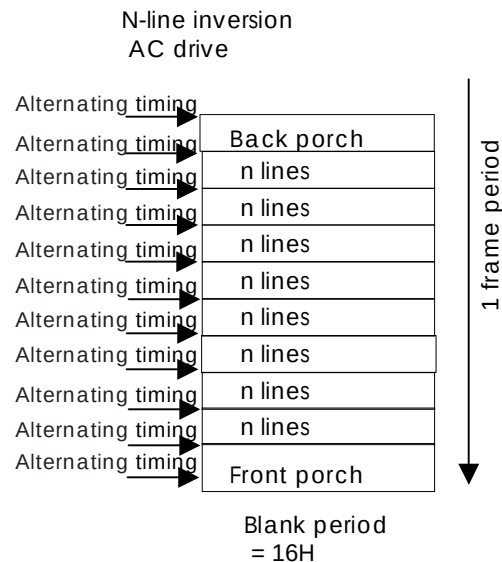
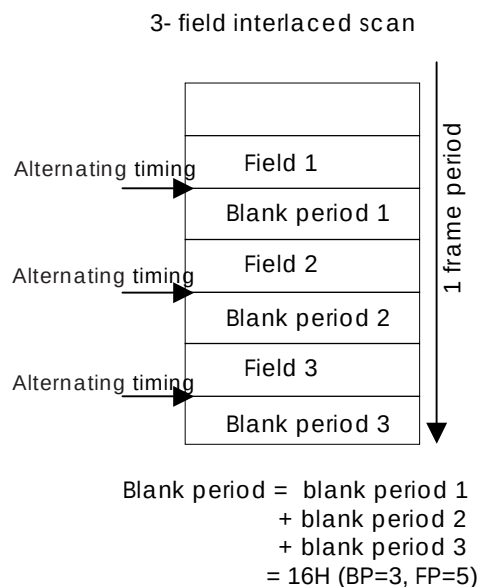
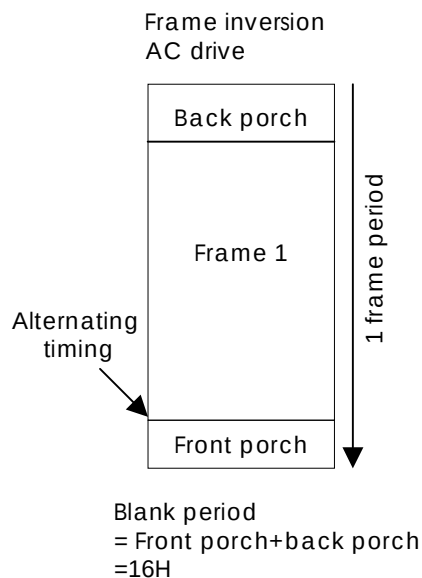
Standby and Sleep Mode



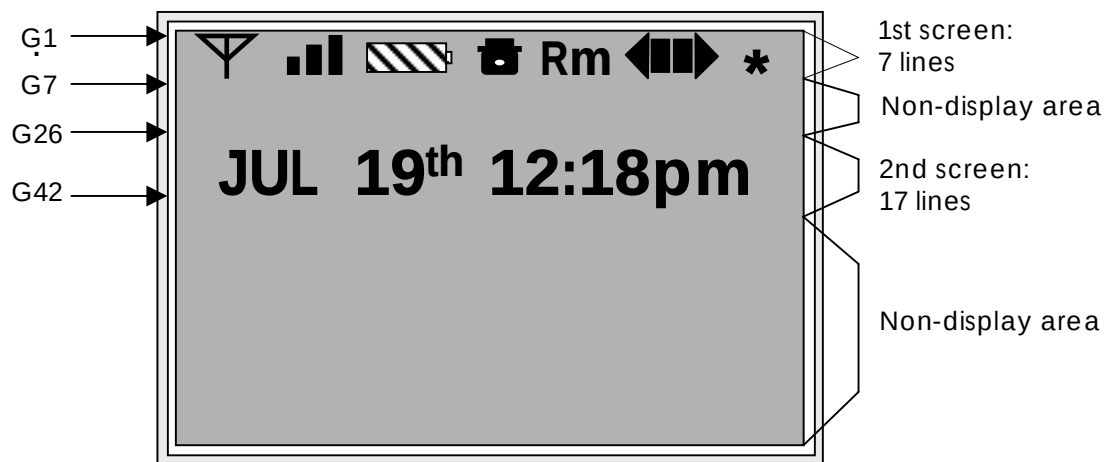
Standby → Shutdown all circuits (SRAM data and control registers basically hold) and only the standby-exit commands can be accepted.

Sleep → RC oscillator, stepup circuits, GVDD/VCOM amplifiers work. Source/gate operations and related commands are not accepted.

Alternating Timing



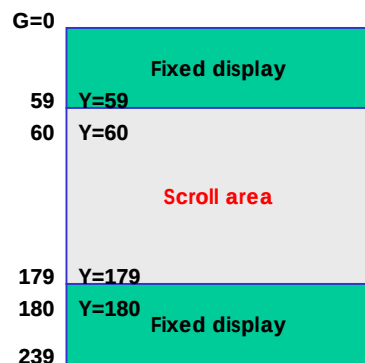
Partial Display



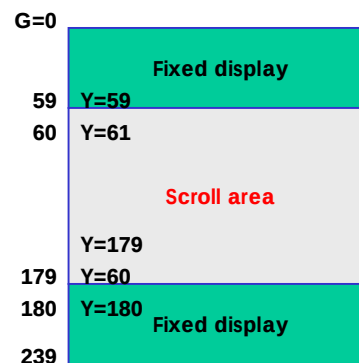
Non-Display Area →

- . Interval Scan
- . No Scan
- . Gate Scan but Source Always GND or Source Following VCOM

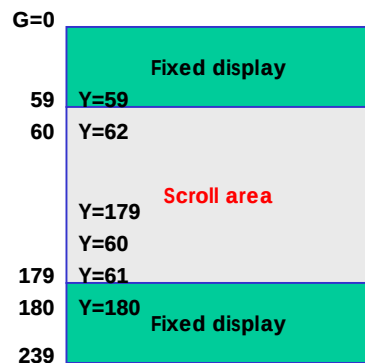
Vertical Scroll Function



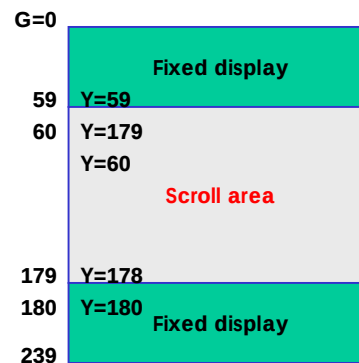
VL=0



VL=1

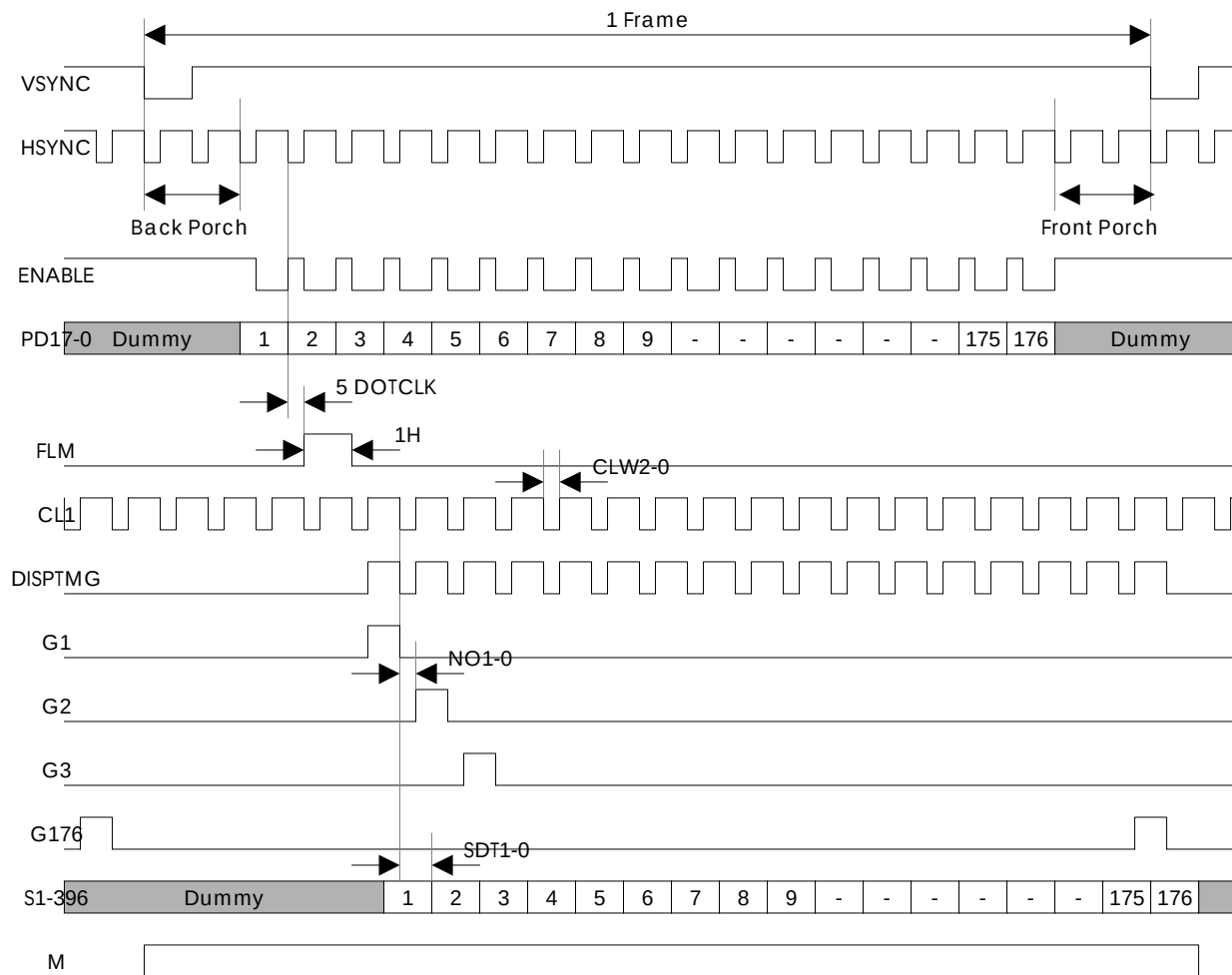


VL=2



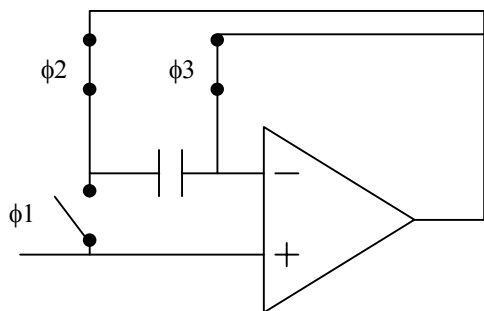
VL=119

Display Timing Control

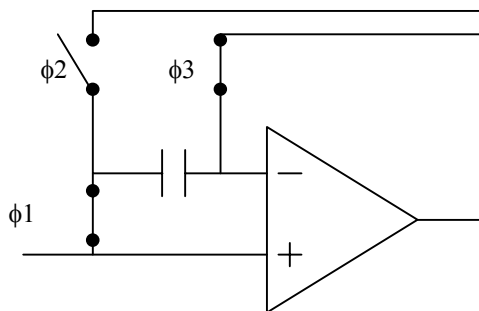


Example of a RGB Interface

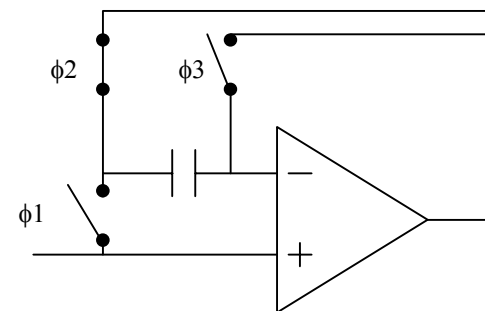
Offset Correction Scheme



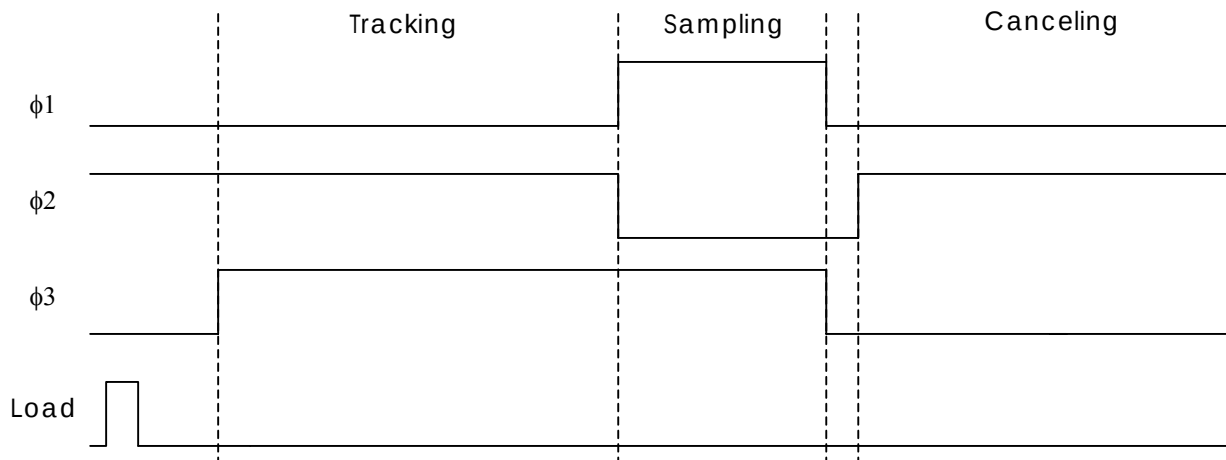
Tracking



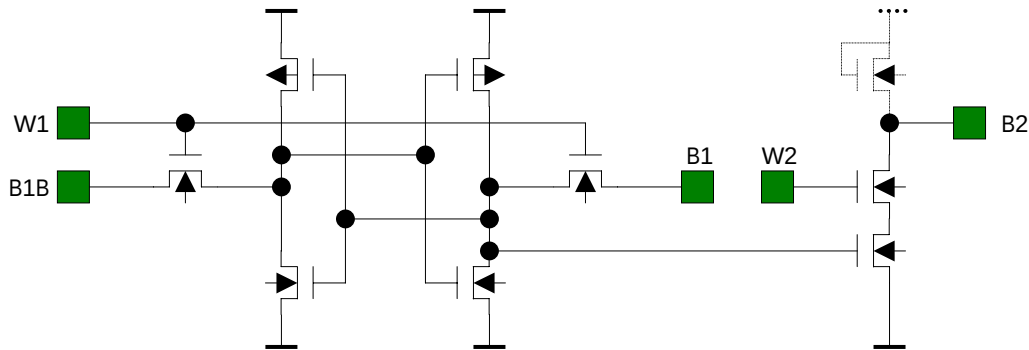
Sampling



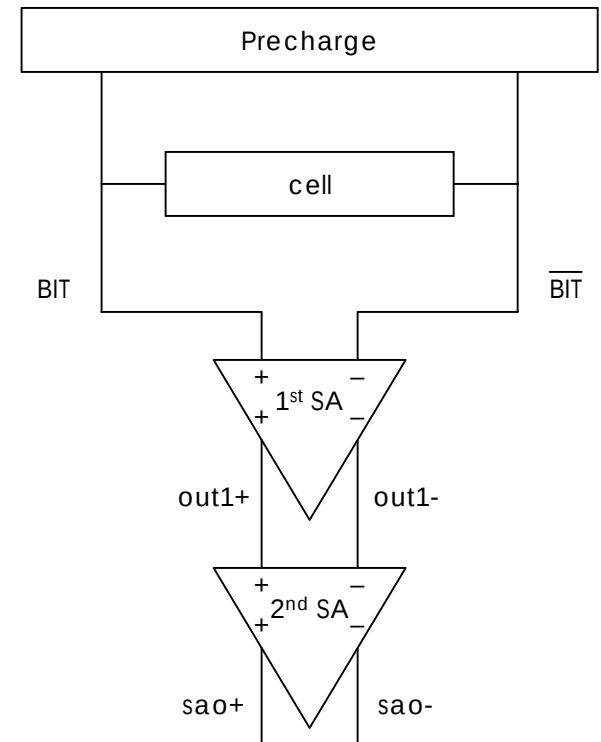
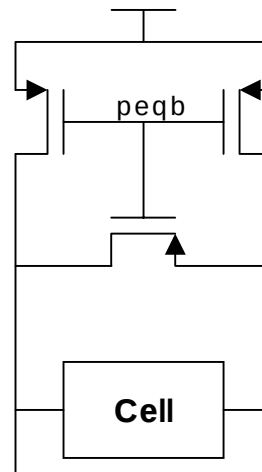
Canceling



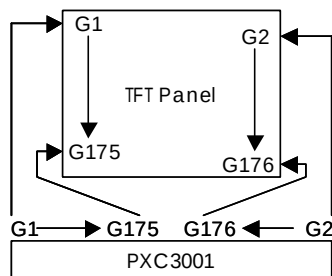
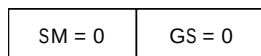
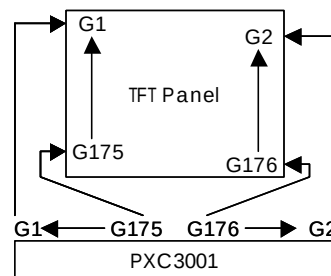
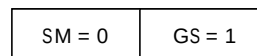
Dual Port SRAM Cell Architecture



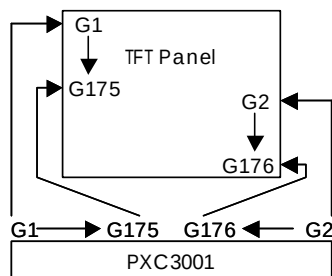
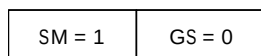
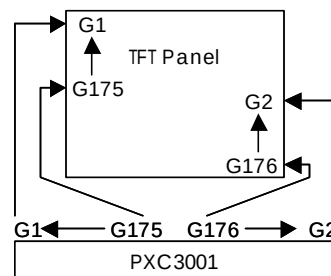
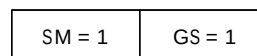
One Read/Write and One Read Only Port



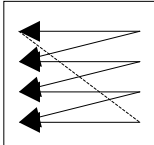
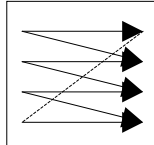
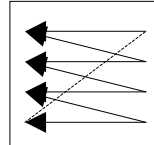
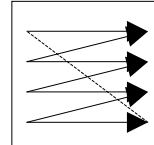
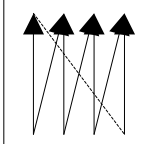
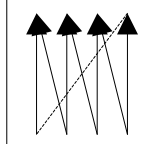
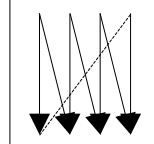
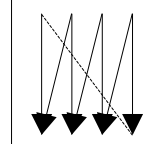
Gate Scan Mode Setting


$$G_1 \rightarrow G_2 \rightarrow G_3 \rightarrow G_4 \rightarrow \dots \rightarrow G_{173} \rightarrow G_{174} \rightarrow G_{175} \rightarrow G_{176}$$


G176 → G175 → G174 → G173
→ ... → G4 → G3 → G2 → G1

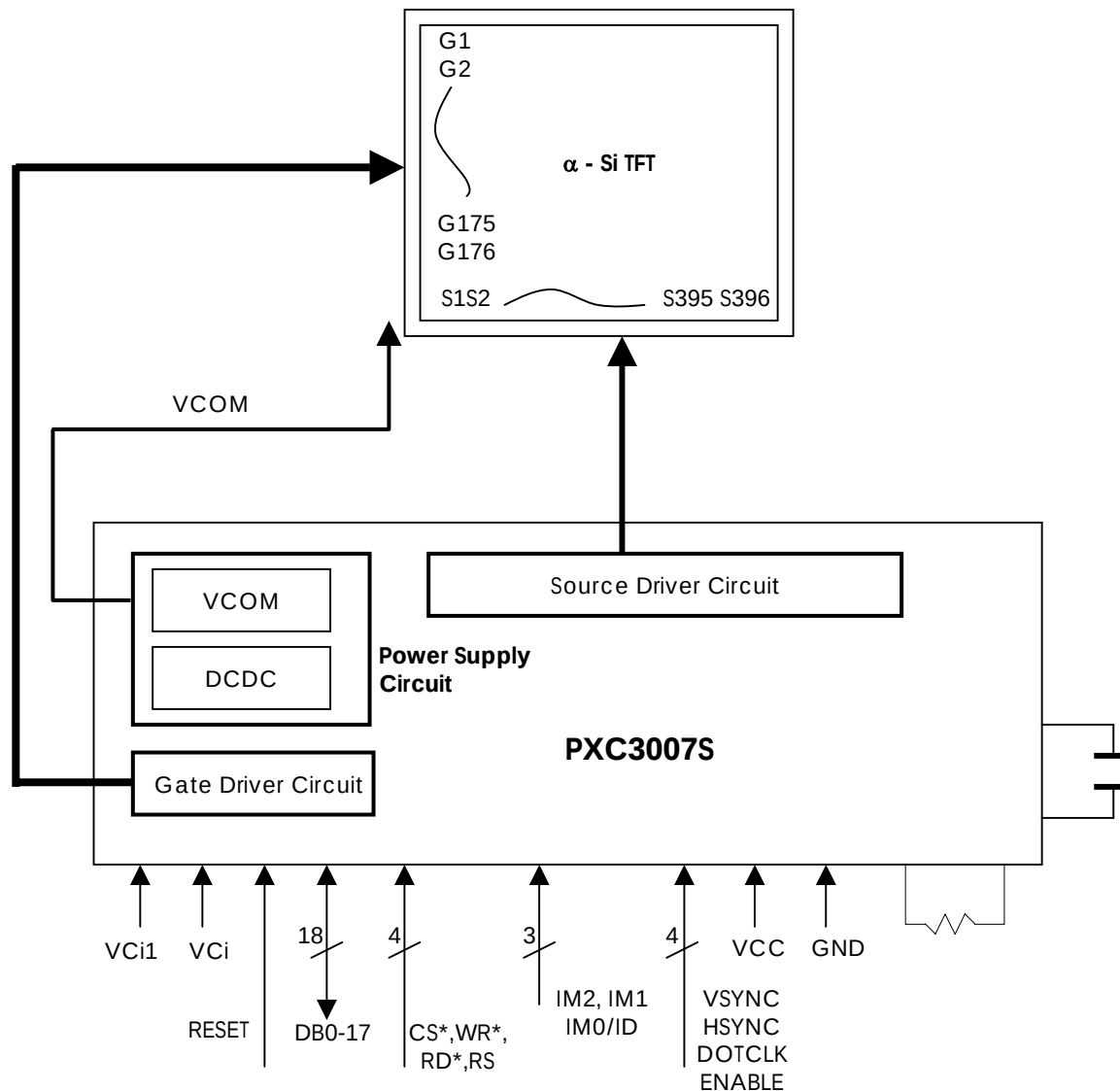

$$G_1 \rightarrow G_3 \rightarrow \dots \rightarrow G_{173} \rightarrow G_{175} \\ \rightarrow G_2 \rightarrow G_4 \rightarrow \dots \rightarrow G_{176}$$

$$G_{176} \rightarrow G_{174} \rightarrow \dots \rightarrow G_4 \rightarrow G_2 \rightarrow G_{175} \rightarrow G_{173} \rightarrow \dots \rightarrow G_1$$

Display Data Entry Mode

	I/D1-0 = "00" Horizontal: dec Vertical: dec	I/D1-0 = "01" Horizontal: inc Vertical: dec	I/D1-0 = "10" Horizontal: dec Vertical: inc	I/D1-0 = "11" Horizontal: inc Vertical: inc
AM=0 Horizontal	0000h  EFAFh	0000h  EFAFh	0000h  EFAFh	0000h  EFAFh
AM=1 Vertical	0000h  EFAFh	0000h  EFAFh	0000h  EFAFh	0000h  EFAFh

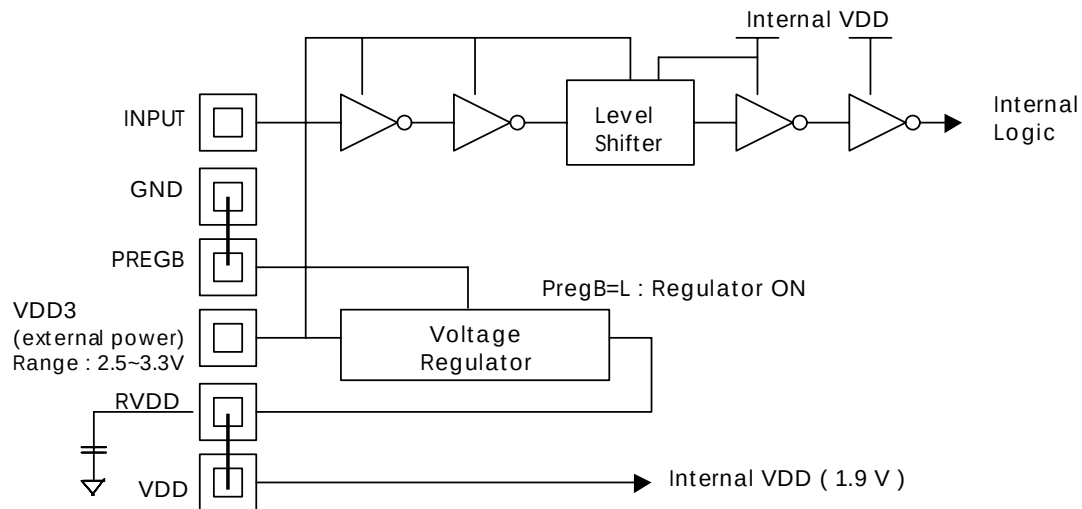
Note : By making a window address area on the internal RAM, it is possible to limit the area to write data.

Mobile LCM System Configuration

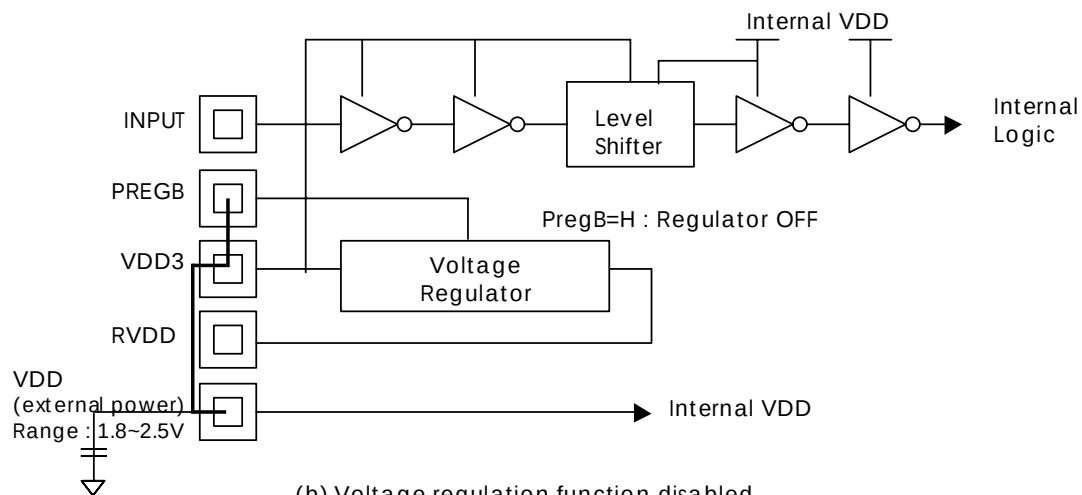




Voltage Regulating Function

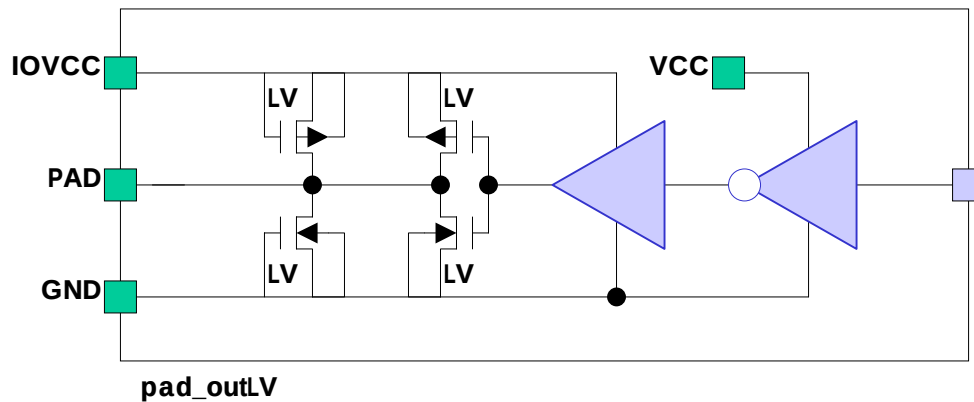
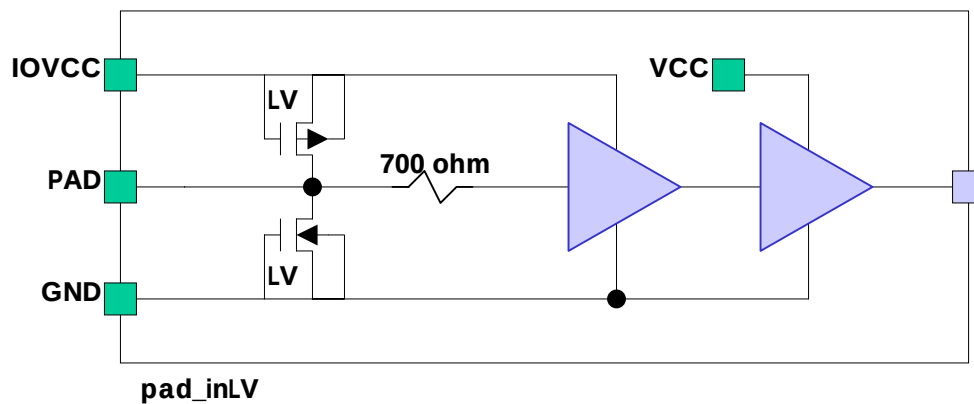


(a) Voltage regulation function enabled



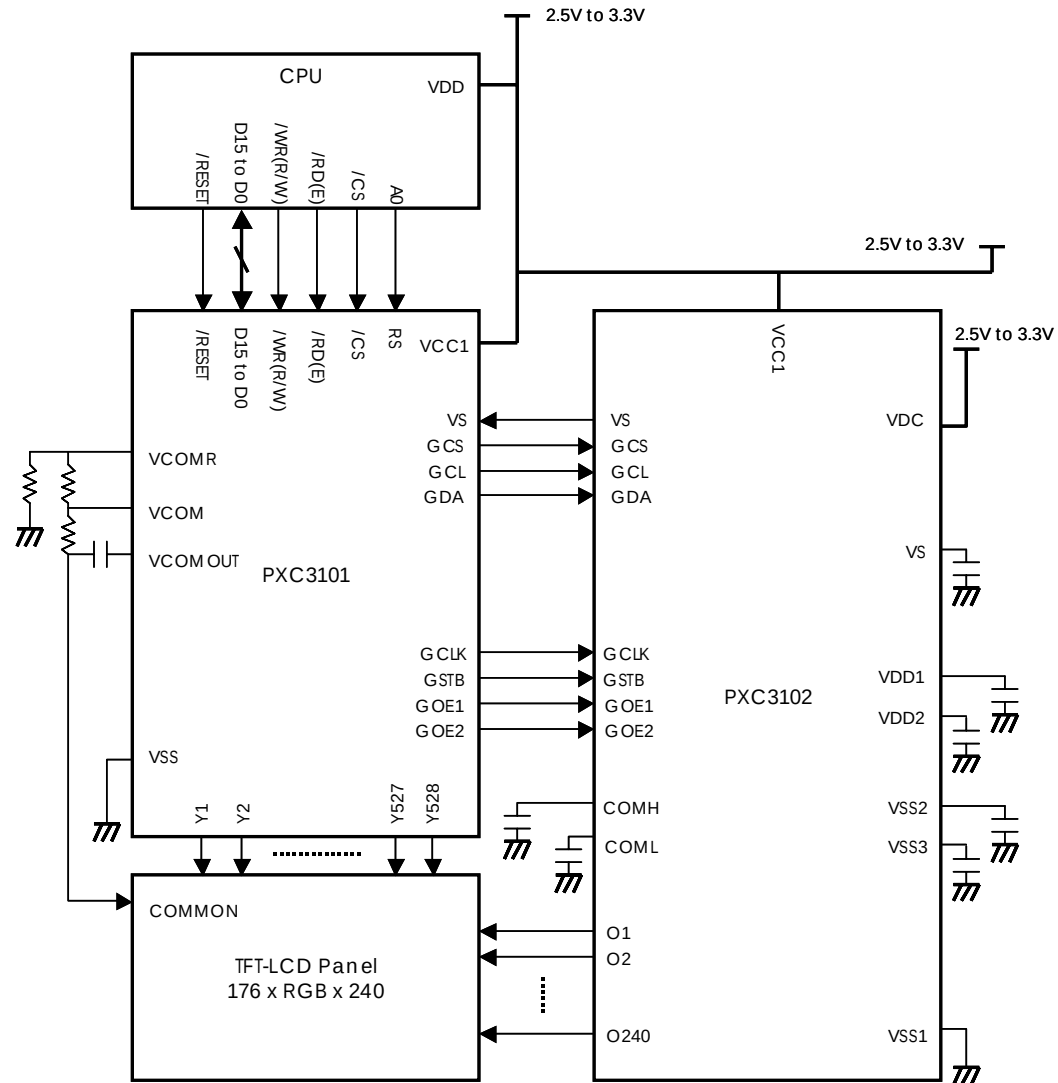
(b) Voltage regulation function disabled

Power for Interface Data Pins

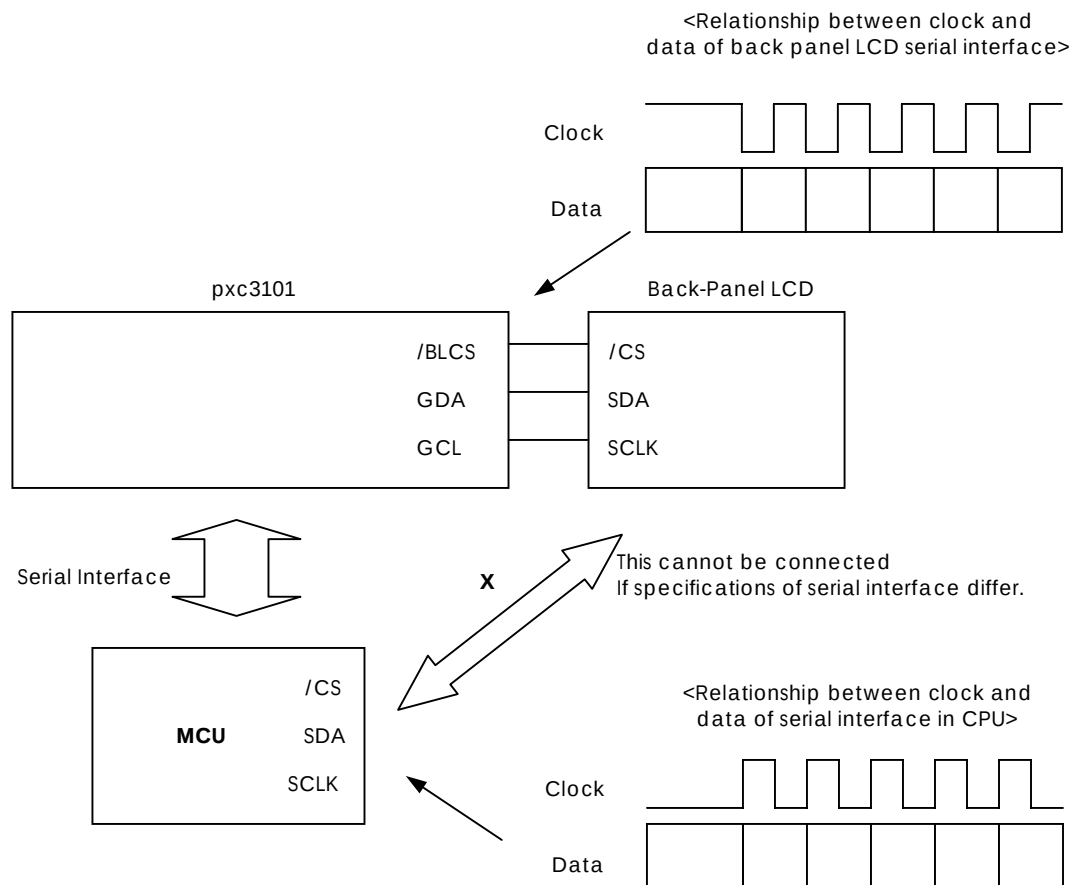


VCC = 2.4V ~ 3.0V
IOVCC = 1.65V ~ 3.0V

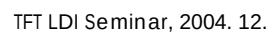
Two-Chip Solution for Mobile TFT



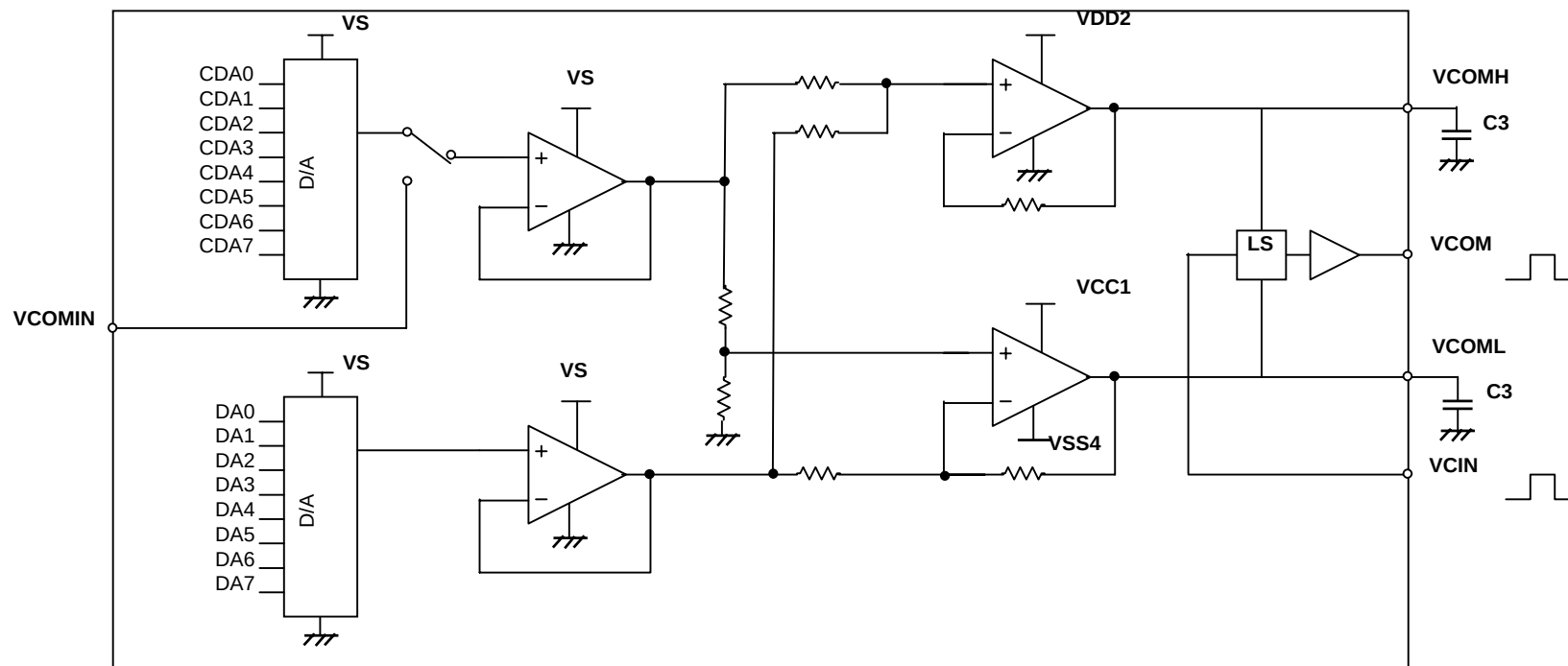
Back Panel Serial Interface thru PXC3101



PixelChips
(주)픽셀칩스



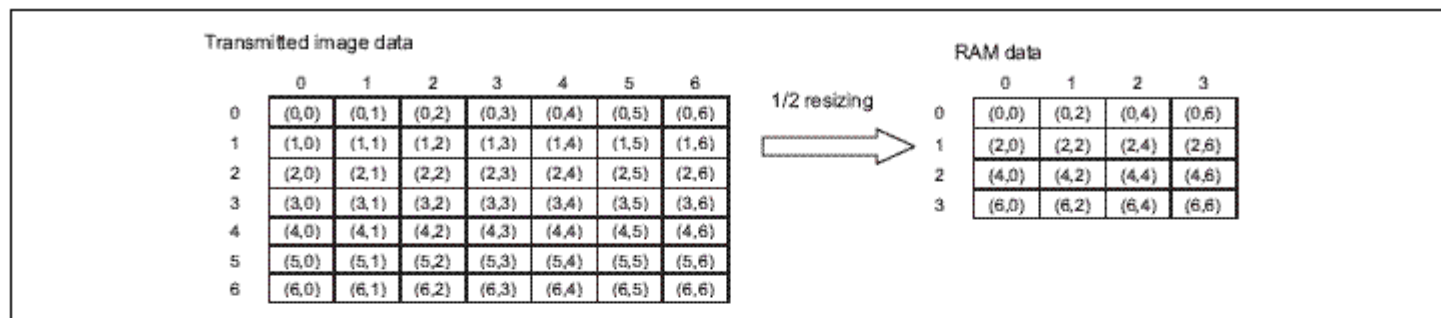
VCOM Amp Circuit 2



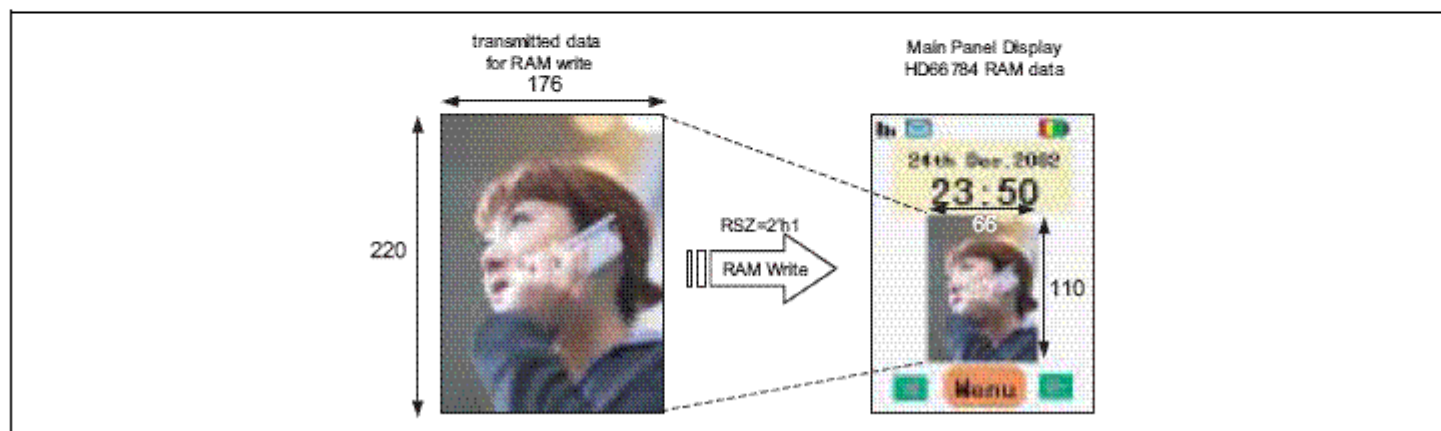
Pixel Chips
(주)픽셀칩스



Resizing

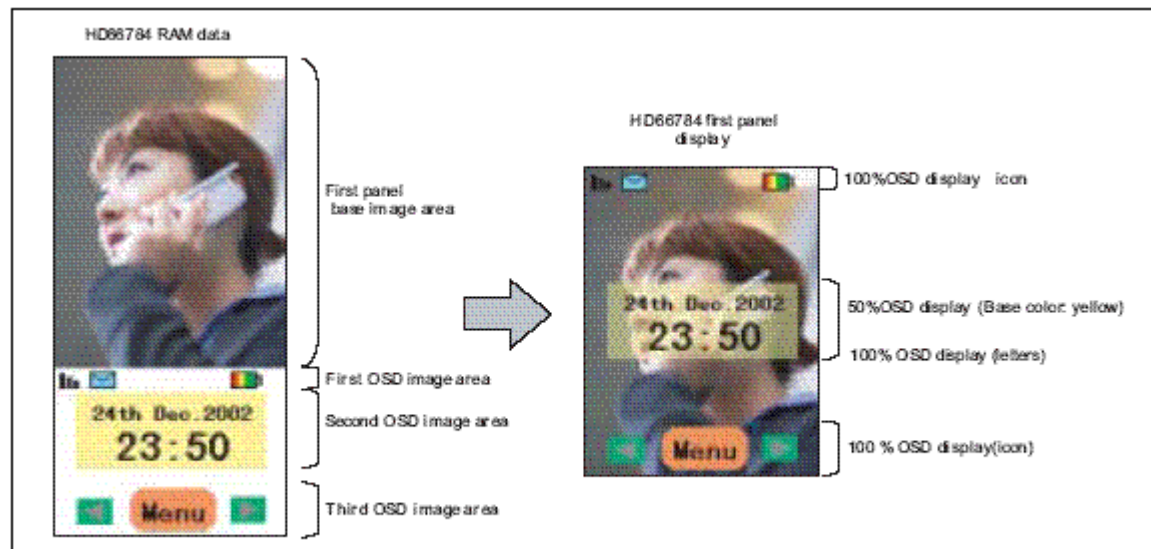


Resizing



Resizing transmission, display example

On Screen Display



Display with OSD and α blending



TFT LDI Seminar, 2004. 12.

Pixel Chips
(주)픽셀칩스



Gate-Less Architecture

